

REGULAR MODELS OF RAMIFIED UNITARY SHIMURA VARIETIES AT MAXIMAL PARAHORIC LEVEL

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ABSTRACT. We use the idea of splitting models to construct and study a semi-stable model for unitary Shimura varieties of signature $(\text{🦋} - \text{🌱}, \text{🌱})$ with maximal parahoric level structure at ramified primes. In this setting, the “naive” splitting model defined by Pappas and Rapoport fails to be flat in a crucial way. We show that the “genuine” splitting model in this case is flat with semi-stable reduction. In addition, we study special cycles within these splitting models.

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1. INTRODUCTION

1.1. Motivation. In the recent paper [Luo25], the second author studies integral models of unitary Shimura varieties of signature $(\text{🦋} - \text{🌱}, \text{🌱})$ with any parahoric level structure at ramified primes. The integral model described in loc.cit. is canonical, meaning it can be characterized using purely group-theoretical data. A moduli interpretation of this “canonical” model was proposed through a conjecture of Smithling in [Smi15], which builds on the work of Pappas and Rapoport in [PR09].

Although the “canonical” integral model possesses many nice properties such as flatness and normality, it is not regular in many important examples. In this paper, we contribute to the construction of regular integral models of Shimura varieties. Our approach is inspired by the *splitting model* first introduced by Pappas and Rapoport in [PR05]. Roughly speaking, the splitting model is a certain modification of the canonical integral model which “refines” the ramification information by adding additional data. Such construction has been studied in the context of Weil restriction [PR05, BH23, SZ25], triality groups [Zha21b], and ramified unitary groups [Krä03, BH23, ZZ25b]. Splitting models have found applications in various settings, including [RX17, Sas19, DK23, BHK⁺20, HLSY23, BH23, BH25, BZZ25], since the first draft of this paper, the splitting model studied here has been used in [LRZ25] for the arithmetic transfer conjecture.

In this paper, we use similar ideas to define and study regular integral models for unitary Shimura varieties with maximal parahoric level at a ramified prime. We give now more details in the following.

1.2. Canonical local model. It is well-known that local properties of integral models can often be studied via local models, which have the advantage of being defined in terms of linear algebra. We adopt this perspective in the present work: we first construct a regular local model and then, in the final section, indicate its implications for the integral models of Rapoport–Zink spaces.

We begin by describing the setup. Let $\mathbb{F}/\mathbb{F}_0$ be a ramified quadratic extension of local fields with residue field $\mathbb{k}$, such that $\text{char}(\mathbb{k}) \neq 2$. Let $\varpi \in \mathbb{F}$ and $\varpi_0 \in \mathbb{F}_0$ be the uniformizers such that $\varpi = \varpi_0 \varpi_1$. Consider a hermitian space $(V, \langle \cdot, \cdot \rangle)$ over $\mathbb{F}$. We are interested in the local model associated to the unitary group $U(V) = \text{Aut}_{\mathbb{F}}(V, \langle \cdot, \cdot \rangle)$.

A *vertex lattice* is a hermitian lattice $\Lambda \subset V$ such that $\varpi_1 \Lambda \subseteq \Lambda \subseteq \varpi \Lambda$, where $\varpi \Lambda$ is the hermitian dual of Λ . The quotient $\varpi \Lambda / \Lambda$ is always of even dimension, say $2n$, called the *type* of the vertex lattice. In the introduction, for simplicity, we assume $n \geq 1$ and take the signature $(n, n) = (n - n, n)$ ¹, it corresponds to the geometric conjugacy class of a certain minuscule cocharacter of $U(V)$, see [PR09, §1.3]. In §2, we will also consider more general signatures.

The *canonical local model* $\text{LM}_{[j]}(\varpi - \varpi_1, \varpi_1)$ is a projective scheme over $\mathbb{A}^1_{\mathbb{k}}$, representing the moduli functor that assigns to each $\mathbb{A}^1_{\mathbb{k}}$ -algebra R the set of filtrations of $\varpi \Lambda \otimes_R \varpi_1 \Lambda$ $\mathbb{F}_0$ -modules

$$\begin{array}{ccccc} \varpi \Lambda \otimes \varpi_1 \Lambda & \longrightarrow & \varpi \Lambda \otimes \varpi_1 \Lambda & \xrightarrow{\varpi_1} & \varpi \Lambda \otimes \varpi_1 \Lambda \\ \uparrow & & \uparrow & & \uparrow \\ \varpi \Lambda & \longrightarrow & \varpi \Lambda & \longrightarrow & \varpi \Lambda \end{array},$$

where $\varpi \Lambda$ and $\varpi_1 \Lambda$ are required to be $\mathbb{F}_0$ -direct summands of rank n , satisfying some additional conditions, see §2.2 for a detailed discussion. The geometry of ramified unitary local models has been studied by various authors, for example [Pap00, PR09, Arz09, Smi15, Yu19, Luo25].

We say that ϖ_1 is ϖ -*modular* when $\varpi = \varpi_1 \varpi_2$ and $\varpi_1 = \varpi_2 \varpi_3$, see Remark 2.1.1 for the terminology used in other cases. The name reflects the fact that the corresponding vertex lattice is ϖ -modular.

Theorem 1.2.1 ([Luo25]). *Assume that ϖ_1 is not ϖ -modular.*

- (i) *The local model $\text{LM}_{[j]}(\varpi - \varpi_1, \varpi_1)$ is flat, normal and Cohen-Macaulay of dimension n .*
- (ii) *The special fiber $\text{LM}_{[j], 0}(\varpi - \varpi_1, \varpi_1)$ is a reduced scheme whose irreducible components are all normal and Cohen-Macaulay. The intersection of n distinct irreducible components has dimension $n - n$.*
- (iii) *There exists a special point $\varpi_1 \in \text{LM}_{[j], 0}(\varpi - \varpi_1, \varpi_1)$, referred to as the worst point. It is the only singular point of each irreducible component of the special fiber and of their intersections.*

1.3. Splitting model. We now turn to the splitting model. Let $\varpi := \varpi_1 \otimes \varpi_2$ and $\varpi_1 := \varpi_2 \otimes \varpi_3$ be the respective elements in $\mathbb{A}^1_{\mathbb{k}} \otimes \mathbb{A}^1_{\mathbb{k}}$. The *naive splitting model* $\text{LM}_{[j]}(\varpi - \varpi_1, \varpi_1)$ is a projective scheme over $\mathbb{A}^1_{\mathbb{k}}$, which represents the moduli functor which sends each $\mathbb{A}^1_{\mathbb{k}}$ -algebra R to the set of filtrations of $\varpi \Lambda \otimes \varpi_1 \Lambda$ $\mathbb{F}_0$ -modules

$$\begin{array}{ccccc} \varpi \Lambda \otimes \varpi_1 \Lambda & \longrightarrow & \varpi \Lambda \otimes \varpi_1 \Lambda & \xrightarrow{\varpi_1} & \varpi \Lambda \otimes \varpi_1 \Lambda \\ \uparrow & & \uparrow & & \uparrow \\ \varpi \Lambda & \longrightarrow & \varpi \Lambda & \longrightarrow & \varpi \Lambda \\ \uparrow & & \uparrow & & \uparrow \\ \varpi \Lambda & \longrightarrow & \varpi \Lambda & \longrightarrow & \varpi \Lambda \end{array}, \quad (1.3.1)$$

such that $(\varpi \Lambda, \varpi_1 \Lambda)$ defines an $\mathbb{F}_0$ -point of the local model $\text{LM}_{[j]}(\varpi - \varpi_1, \varpi_1)$, and for each $\varpi \in \{\varpi_1, \varpi_2\}$, ϖ_1 is an $\mathbb{F}_0$ -direct summand of rank n of $\varpi \Lambda$ which satisfies the Krämer condition:

$$(\varpi + \varpi_1)(\varpi_1) \subseteq \varpi_1, \quad (\varpi - \varpi_1)(\varpi_1) = (0). \quad (1.3.2)$$

¹when $\varpi = \varpi_1$, the same results apply, but the reflex field of the canonical local model is $\mathbb{F}_0$.

We refer to this as the *naive splitting model*, as it generally fails to be flat.

The *splitting model* $\mathfrak{c}_{[\mathfrak{p}]}(\mathfrak{b} - \mathfrak{g}, \mathfrak{g})$ is defined as the scheme-theoretic closure of its generic fiber $\mathfrak{c}_{[\mathfrak{p}]}(\mathfrak{b}, \mathfrak{g})$ in $\mathfrak{c}_{[\mathfrak{p}]}(\mathfrak{b} - \mathfrak{g}, \mathfrak{g})$. By construction, we have a natural forgetful map

$$\mathfrak{f} : \mathfrak{c}_{[\mathfrak{p}]}(\mathfrak{b} - \mathfrak{g}, \mathfrak{g}) \longrightarrow \mathfrak{c}_{[\mathfrak{p}]}(\mathfrak{b} - \mathfrak{g}, \mathfrak{g}).$$

Suppose now that the special fiber $\mathfrak{c}_{[\mathfrak{p}]}(\mathfrak{b} - \mathfrak{g}, \mathfrak{g})$ has $\mathfrak{r} = \mathfrak{r}(\mathfrak{b}, \mathfrak{g})$ irreducible components, see Theorem 2.2.2 for the precise value of $\mathfrak{r}$ in each case. Denote the irreducible components as $\mathfrak{f}_i$ for $\mathfrak{g} \leq \mathfrak{f}_i \leq \mathfrak{r}$. The first main result of the paper is:

Theorem 1.3.1 (cf. §3.4.5). *Assume that $\mathfrak{b} \geq \mathfrak{g}$ and that $\mathfrak{g}$ is not $\mathfrak{p}$ -modular.*

- (i) *The splitting model $\mathfrak{c}_{[\mathfrak{p}]}(\mathfrak{b} - \mathfrak{g}, \mathfrak{g})$ is flat of dimension $\mathfrak{b}$, with semi-stable reduction.*
- (ii) *The special fiber of the splitting model has $\mathfrak{r} + \mathfrak{g}$ irreducible components: $\mathfrak{f}_i$ for $\mathfrak{g} \leq \mathfrak{f}_i \leq \mathfrak{r}$ and $\mathfrak{f}_0$. All components are smooth of dimension $\mathfrak{b} - \mathfrak{g}$.*
- (iii) *The forgetful map $\mathfrak{f} : \mathfrak{c}_{[\mathfrak{p}]}(\mathfrak{b} - \mathfrak{g}, \mathfrak{g}) \rightarrow \mathfrak{c}_{[\mathfrak{p}]}(\mathfrak{b} - \mathfrak{g}, \mathfrak{g})$ maps $\mathfrak{f}_i$ to $\mathfrak{f}_i$ for $\mathfrak{g} \leq \mathfrak{f}_i \leq \mathfrak{r}$, and $\mathfrak{f}_0$ to the worst point $\mathfrak{f}_0$.*

It is straightforward to verify that $\mathfrak{f}$ is an isomorphism away from the worst point. As a consequence, we obtain the following:

Corollary 1.3.2. *When $\mathfrak{g}$ is not $\mathfrak{p}$ -modular, the open subscheme $\mathfrak{c}_{[\mathfrak{p}]}(\mathfrak{b} - \mathfrak{g}, \mathfrak{g}) \setminus \{\mathfrak{f}_0\}$ of the canonical local model has semi-stable reduction.*

This leads one to ask whether the splitting model is in fact a blow-up of the canonical model. Indeed, we managed to characterize the splitting model in this way:

Theorem 1.3.3 (Thm. 4.0.1). *Assume $\mathfrak{g}$ is not $\mathfrak{p}$ -modular. The map $\mathfrak{f} : \mathfrak{c}_{[\mathfrak{p}]}(\mathfrak{b} - \mathfrak{g}, \mathfrak{g}) \rightarrow \mathfrak{c}_{[\mathfrak{p}]}(\mathfrak{b} - \mathfrak{g}, \mathfrak{g})$ is the blow-up of $\mathfrak{c}_{[\mathfrak{p}]}(\mathfrak{b} - \mathfrak{g}, \mathfrak{g})$ along the worst point. Under this identification, $\mathfrak{f}_0$ is the exceptional divisor of the blow-up, and $\mathfrak{f}_i$ is the strict transform of $\mathfrak{f}_i$ for $\mathfrak{g} \leq \mathfrak{f}_i \leq \mathfrak{r}$.*

Remark 1.3.4. When $\mathfrak{g}$ is $\mathfrak{p}$ -modular, the canonical local model is smooth and connected. In contrast to other cases, it does not exhibit an analogue of the worst point. Moreover, in this case, the splitting model coincides with the canonical local model, see also Remark 2.5.3.

Characterizing the splitting model as a blow-up certainly suggests that it is the “canonical” choice in some sense, see also [Gor23]. It remains an interesting open question to justify this construction from a group-theoretic perspective.

1.4. Signature conditions. Our definition of $\mathfrak{c}_{[\mathfrak{p}]}(\mathfrak{b} - \mathfrak{g}, \mathfrak{g})$ requires that the pair $(\mathfrak{b}, \mathfrak{g})$ defines an $\mathfrak{p}$ -point of the canonical local model. In practice, characterizing the points of the canonical local model is a subtle problem; we refer the reader to the introduction of [Luo25] for further discussion. In our setting, $\mathfrak{c}_{[\mathfrak{p}]}(\mathfrak{b} - \mathfrak{g}, \mathfrak{g})$ admits a moduli description via the *strengthened spin condition* introduced in [Smi15]. However, this condition is intricate and often difficult to verify in applications. We prove the following:

Theorem 1.4.1 (Thm. 3.3.4). *When $(\mathfrak{b}, \mathfrak{g}) = (\mathfrak{b} - \mathfrak{g}, \mathfrak{g})$ and $\mathfrak{g}$ is not $\mathfrak{p}$ -modular, the Krämer condition implies the strengthened spin condition.*

Therefore, we can drop the strengthened spin condition when defining the splitting model, see §1.5 or §5.2. Consequently, we obtain an alternative definition of the naive splitting model, which proves useful in practice, see §1.6 below.

Remarks 1.4.2. If we drop the strengthened spin condition when defining the naive splitting model in the very beginning, it is not clear that we have a natural forgetful functor $\mathfrak{f} : \mathfrak{c}_{[\mathfrak{p}]}(\mathfrak{b} - \mathfrak{g}, \mathfrak{g}) \rightarrow \mathfrak{c}_{[\mathfrak{p}]}(\mathfrak{b} - \mathfrak{g}, \mathfrak{g})$. But this is clear within the strengthened spin condition. Also, it is not clear to us whether Proposition 1.4.1 holds for general signature $(\mathfrak{b}, \mathfrak{g})$.

1.5. Special cycles on Rapoport–Zink spaces. Although the construction of $\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}} - \mathcal{H}^{\text{ss}}, \mathcal{H}^{\text{ss}})$ via scheme-theoretic closure obscures its moduli interpretation, we are nevertheless able to define two classes of special cycles on the associated splitting Rapoport–Zink (RZ) space and to verify certain useful properties that do not hold in the canonical RZ space.

Fix a basic framing object $\mathcal{H}^{\text{ss}}$ of dimension $\dim \mathcal{H}^{\text{ss}}$ and type $\text{type}(\mathcal{H}^{\text{ss}})$ over $\mathcal{H}^{\text{ss}}$, where $\mathcal{H}^{\text{ss}}$ is the residue field of $\mathcal{H}^{\text{ss}}$ and $\mathcal{H}^{\text{ss}} = \overline{\mathcal{H}^{\text{ss}}}$. See §5 for the precise definition. Associate to such a framing object is the *canonical RZ space* $\mathcal{H}^{\text{ss}}$ over $\mathcal{H}^{\text{ss}}$. The *naive splitting RZ space* $\mathcal{H}^{\text{ss}}$ is defined over $\mathcal{H}^{\text{ss}}$ by classifying additional splitting structures. We then define the *splitting RZ space* $\mathcal{H}^{\text{ss}}$ as the flat closure of $\mathcal{H}^{\text{ss}}$. For precise definitions, we refer the reader to §5.2. The terminology is motivated by the corresponding local model constructions.

Let $\mathcal{H}^{\text{ss}}$ be the unique Lubin–Tate $\mathcal{H}^{\text{ss}}$ -module over $\mathcal{H}^{\text{ss}}$ and let $\mathcal{H}^{\text{ss}}$ be its canonical lifting to $\mathcal{H}^{\text{ss}}$. Define $\mathcal{H}^{\text{ss}} = \mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}}, \mathcal{H}^{\text{ss}}) \otimes \mathcal{H}^{\text{ss}}$ to be the space of special quasi-homomorphisms of the framing objects. It carries a natural hermitian form $\langle \cdot, \cdot \rangle$. For any $\mathcal{H}^{\text{ss}} \in \mathcal{H}^{\text{ss}}$, we define the *Kudla–Rapoport special cycles* $\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})$ and $\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})$ as closed formal subschemes of $\mathcal{H}^{\text{ss}}$. We define the *Kudla–Rapoport special cycles* $\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}}$ and $\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}}$ on $\mathcal{H}^{\text{ss}}$ as the pull-back along the natural projection $\mathcal{H}^{\text{ss}} \rightarrow \mathcal{H}^{\text{ss}}$. We refer the reader to Definition 5.4.1 for the precise formulation. We prove the following:

Proposition 1.5.1 (Prop. 5.4.4). *Let $\mathcal{H}^{\text{ss}} \in \mathcal{H}^{\text{ss}}$ be any nonzero vector. If the closed formal subschemes $\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}} \subseteq \mathcal{H}^{\text{ss}}$ and $\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}} \subseteq \mathcal{H}^{\text{ss}}$ are nonempty, then they are Cartier divisors.*

A key step in the proof of Proposition 1.5.1 is to relate special cycles to sections of the *line bundles of modular forms* $\mathcal{H}^{\text{ss}}$ (resp. $\mathcal{H}^{\text{ss}}$) in the square-zero extension of special cycles.

Remark 1.5.2. (i) See also [LRZ25, Theorem 6.1.3] for an extreme example in which special cycles on $\mathcal{H}^{\text{ss}}$ fail to be Cartier divisors. In the forthcoming work [LSR26], we will show that, for most maximal parahoric level structures at ramified places, special cycles are likewise not Cartier divisors.

(ii) The nonemptiness of the special divisor can be explicitly checked by Bruhat–Tits stratification for reduced locus of the RZ space or the splitting RZ space studied in [HLS26, ZZ25a], using arguments similar to those of [HSY23, HLSY23].

By Proposition 1.5.1, when $\mathcal{H}^{\text{ss}} \neq \mathcal{H}^{\text{ss}}$, the special divisors $\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}}$ and $\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}}$ define classes $[\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}}]$ and $[\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}}]$ in the Grothendieck group $\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})$. We refer the reader to §5.4 for the definition of the $\mathcal{H}^{\text{ss}}$ -group. We define $[\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}}]$ resp. $[\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}}]$ as the classes in $\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})$ of the following two-term complexes of $\mathcal{H}^{\text{ss}}$ -modules concentrated in degrees $-\mathcal{H}^{\text{ss}}$ and $\mathcal{H}^{\text{ss}}$:

$$(\cdots \rightarrow \mathcal{H}^{\text{ss}} \rightarrow \mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}} \rightarrow \mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}} \rightarrow \mathcal{H}^{\text{ss}}), \quad \text{resp.} \quad (\cdots \rightarrow \mathcal{H}^{\text{ss}} \rightarrow \mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}} \rightarrow \mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}} \rightarrow \mathcal{H}^{\text{ss}}).$$

We prove the linear invariance of the intersection of special divisors, which generalizes the main result of [How19].

Proposition 1.5.3. *Assume that $\mathcal{H}^{\text{ss}}, \dots, \mathcal{H}^{\text{ss}} \in \mathcal{H}^{\text{ss}}$ and $\mathcal{H}^{\text{ss}}, \dots, \mathcal{H}^{\text{ss}} \in \mathcal{H}^{\text{ss}}$ generate the same $\mathcal{H}^{\text{ss}}$ -submodule. Then we have*

$$[\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}}] \cdots [\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}}] = [\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}}] \cdots [\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}}] \in \mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}}).$$

Similarly, we have

$$[\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}}] \cdots [\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}}] = [\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}}] \cdots [\mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}})^{\text{ss}}] \in \mathcal{H}^{\text{ss}}(\mathcal{H}^{\text{ss}}).$$

1.6. Comments. The main case we consider is the *strongly non-special* case, see Remark 2.1.1. This case exhibits the worst ramification compared to earlier works. In this setting, the “exceptional divisor” of the naive splitting model has two irreducible components, one of which does not lift to the generic fiber. An interesting direction for future research is to study the differences between these two components and to

identify conditions that distinguish them. This may eventually lead to a moduli-theoretic characterization of the splitting model.

We also mention a recent paper by Zachos and Zhao [ZZ25c], where they introduce a different splitting local model $\mathcal{M}_{[j]}(\mathfrak{g}, \mathfrak{h})$ defined by adding the additional filtration $\mathfrak{g}_{[j]}(\mathfrak{h})$, (but not $\mathfrak{g}_{[j]}(\mathfrak{h})$). They further construct $\mathcal{M}_{[j]}(\mathfrak{g}, \mathfrak{h})$ by blowing up the preimage of the worst point in $\mathcal{M}_{[j]}(\mathfrak{g}, \mathfrak{h})$. They show that $\mathcal{M}_{[j]}(\mathfrak{g}, \mathfrak{h})$ and $\mathcal{M}_{[j]}(\mathfrak{g}, \mathfrak{h})$ are flat, normal and Cohen-Macaulay, and that $\mathcal{M}_{[j]}(\mathfrak{g}, \mathfrak{h})$ is semi-stable. In later work, Zachos and Zhao [ZZ25a, Appendix] prove that their $\mathcal{M}_{[j]}(\mathfrak{g}, \mathfrak{h})$ coincides with our splitting model $\mathcal{M}_{[j]}(\mathfrak{g}, \mathfrak{h})$.

However, since their construction introduces filtrations only on one side, the spaces $\mathcal{M}_{[j]}(\mathfrak{g}, \mathfrak{h})$ and $\mathcal{M}_{[j]}(\mathfrak{g}, \mathfrak{h})$ are not isomorphic. In fact, a direct computation shows that $\mathcal{M}_{[j]}(\mathfrak{g}, \mathfrak{h})$ is not flat in general, due to similar reasons underlying the failure of flatness in $\mathcal{M}_{[j]}(\mathfrak{g}, \mathfrak{h})$. It remains an interesting problem to give a more geometric explanation of the failure of flatness and to find a moduli interpretation of the flat integral model.

1.7. The structure of the paper. The organization of the paper is as follows. In Section 2, we define the splitting model for any signature and establish some geometric properties. For the remaining sections, we will assume the signature is $(\mathfrak{g}, \mathfrak{h}) = (\mathfrak{g}, \mathfrak{h})$. In Section 3, we prove that the Krämer condition implies the strengthened spin condition, and the splitting model is semi-stable. In Section 4, we prove that the splitting model is the blow-up of the canonical local model at the worst point. In Section 5, we define and study the corresponding splitting Rapoport–Zink spaces and special cycles. We prove that special cycles are Cartier divisors and that they satisfy the linear invariance property.

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1.9. Notations.

- For any $\mathcal{O}_F$ -algebra R with a structure map $\mathfrak{g} : \mathcal{O}_F \rightarrow R$, we denote $\mathfrak{g} := \mathfrak{g} \otimes \mathfrak{g}$ and $\mathfrak{g} := \mathfrak{g} \otimes \mathfrak{g}$.
- For an integer $0 \leq j \leq n$, we define $\mathfrak{g}_j := \mathfrak{g} + \mathfrak{g} - \mathfrak{g}$.
- Let R be a DVR and let $\mathfrak{g} \rightarrow \mathfrak{g}$ be an arithmetic scheme. We denote by $\mathfrak{g}_{\text{gen}}$ its generic fiber and $\mathfrak{g}_{\text{sp}}$ its special fiber.
- We consider matrices:

$$\mathfrak{g} = \begin{pmatrix} & & \mathfrak{g} \\ & \ddots & \\ \mathfrak{g} & & \end{pmatrix}, \quad \mathfrak{g} = \begin{pmatrix} & \mathfrak{g} \\ -\mathfrak{g} & \end{pmatrix}, \quad \mathfrak{g} = \begin{pmatrix} & \mathfrak{g} \\ -\mathfrak{g} & \mathfrak{g} \end{pmatrix}.$$

For any matrix $\mathfrak{g}$, we define $\mathfrak{g} := \mathfrak{g} \otimes \mathfrak{g}$.

2. MODULI PROBLEMS

2.1. Basic setup. Let $\mathcal{O}_F/\mathcal{O}_F$ be a quadratic extension of complete discrete valuation rings with the same perfect residue field $\mathfrak{f}$ with $\mathfrak{g}(\mathfrak{f}) = \mathfrak{g} \neq \mathfrak{g}$ and uniformizers $\mathfrak{g}$, resp. $\mathfrak{g}$ such that $\mathfrak{g} = \mathfrak{g}$.

Consider the vector space $\mathbb{C}^{\mathcal{B}}$ with a set of $\mathbb{C}$ -basis $\mathbf{e}_1, \dots, \mathbf{e}_n$ which we call the *standard basis*. Let

$$\langle \cdot, \cdot \rangle : \mathbb{C}^{\mathcal{B}} \times \mathbb{C}^{\mathcal{B}} \rightarrow \mathbb{C}$$

be the $\mathbb{C}/\mathbb{C}$ -Hermitian form which is split with respect to the standard basis, that is,

$$\langle \mathbf{e}_i, \mathbf{e}_j \rangle = \delta_{ij}, \quad \mathbf{e}_i, \mathbf{e}_j \in \mathbb{C}, \quad (2.1.1)$$

where $\mathbf{e}_i \mapsto \bar{\mathbf{e}}_i$ is the nontrivial element of $\mathbb{C}/\mathbb{C}$ and $\mathbf{e}_i = \mathbf{e}_i + \mathbf{e}_i - \mathbf{e}_i$. Attached to $\langle \cdot, \cdot \rangle$ are the respective alternating and symmetric $\mathbb{C}$ -bilinear forms

$$\langle \cdot, \cdot \rangle \times \langle \cdot, \cdot \rangle \rightarrow \mathbb{C}$$

given by

$$\langle \mathbf{e}_i, \mathbf{e}_j \rangle := \frac{1}{2} (\mathbf{e}_i - \mathbf{e}_j, \mathbf{e}_i + \mathbf{e}_j) \quad \text{and} \quad (\mathbf{e}_i, \mathbf{e}_j) := \frac{1}{2} (\langle \mathbf{e}_i, \mathbf{e}_j \rangle + \langle \mathbf{e}_j, \mathbf{e}_i \rangle).$$

For each integer $\mathbf{e} = \mathbf{e}_i + \mathbf{e}_j$ with $\mathbf{e} \leq \mathbf{e}_i < \mathbf{e}_j$, define the standard $\mathbb{C}$ -lattice

$$\mathcal{L}_{\mathbf{e}} := \sum_{\mathbf{e} = \mathbf{e}_i + \mathbf{e}_j} \mathbb{C} \mathbf{e}_i + \mathbb{C} \mathbf{e}_j + \sum_{\mathbf{e} = \mathbf{e}_i + \mathbf{e}_j} \mathbb{C} \mathbf{e}_i + \mathbb{C} \mathbf{e}_j \subset \mathbb{C}^{\mathcal{B}}. \quad (2.1.2)$$

For all $\mathbf{e}$, the $\langle \cdot, \cdot \rangle$ -dual of $\mathcal{L}_{\mathbf{e}}$ in $\mathbb{C}^{\mathcal{B}}$ is $\mathcal{L}_{-\mathbf{e}}$, by which we mean

$$\{ \mathbf{e}_i \in \mathbb{C}^{\mathcal{B}} \mid \langle \mathbf{e}_i, \mathbf{e}_j \rangle \in \mathbb{C} \} = \mathcal{L}_{-\mathbf{e}},$$

and

$$\mathcal{L}_{\mathbf{e}} \times \mathcal{L}_{-\mathbf{e}} \xrightarrow{\langle \cdot, \cdot \rangle} \mathbb{C} \quad (2.1.3)$$

is a perfect $\mathbb{C}$ -bilinear pairing. Similarly, $\mathcal{L}_{\mathbf{e}}$ is the $(\cdot, \cdot)$ -dual of $\mathcal{L}_{-\mathbf{e}}$ in $\mathbb{C}^{\mathcal{B}}$. The $\mathcal{L}_{\mathbf{e}}$'s form a complete, periodic, self-dual lattice chain

$$\dots \subset \mathcal{L}_{-\mathbf{e}_i} \subset \mathcal{L}_{-\mathbf{e}_j} \subset \mathcal{L}_{\mathbf{e}_i} \subset \mathcal{L}_{\mathbf{e}_j} \subset \dots$$

For an integer $\mathbf{e} \leq \mathbf{e}_i \leq \mathbf{e}_j$, let $[\mathbf{e}] = \{\pm \mathbf{e}\} + \mathbf{e}_i \in \mathbb{C}$, we will be interested in the sub-lattice chain $\mathcal{L}_{[\mathbf{e}]}$:

$$\dots \subset \mathcal{L}_{-\mathbf{e}_i} \subset \mathcal{L}_{-\mathbf{e}_j} \subset \mathcal{L}_{\mathbf{e}_i} \subset \mathcal{L}_{\mathbf{e}_j} \subset \dots \quad (2.1.4)$$

Remark 2.1.1. We classify the cases according to different choices of $\mathbf{e}$, and assign the following names:

- The case $\mathbf{e} = \mathbf{e}_i$ is called the *self-dual* case.
- The case $\mathbf{e} = \mathbf{e}_i + \mathbf{e}_j$ and $\mathbf{e} = \mathbf{e}_i - \mathbf{e}_j$ is called the *$\mathbb{C}$ -modular* case.
- The case $\mathbf{e} = \mathbf{e}_i + \mathbf{e}_j + \mathbf{e}_k$ and $\mathbf{e} = \mathbf{e}_i - \mathbf{e}_j$ is called the *almost $\mathbb{C}$ -modular* case.
- The case $\mathbf{e} = \mathbf{e}_i + \mathbf{e}_j$ and $\mathbf{e} = \mathbf{e}_i - \mathbf{e}_j$ is called *Yu's* case.
- All other cases are called *strongly non-special*.

2.2. Canonical local model. Fix an integer $\mathbf{e} \leq \mathbf{e}_i \leq \lfloor \frac{\mathbf{e}_i}{2} \rfloor$. We recall the definition of the canonical local model with index $[\mathbf{e}]$. Let $\mathbf{r}, \mathbf{g}$ be natural numbers such that $\mathbf{r} + \mathbf{g} = \mathbf{e}$. Let $\mathbf{e} = \mathbf{e}_i$ when $\mathbf{r} = \mathbf{g}$ and $\mathbf{e} = \mathbf{e}_i$ otherwise.

Definition 2.2.1. (i) The *naive local model* $\mathcal{M}_{[\mathbf{e}]}(\mathbf{r}, \mathbf{g})$ is a projective scheme over $\mathbb{A}^1_{\mathbb{C}}$. It represents the moduli functor that sends each $\mathbb{C}$ -algebra $\mathbb{C}$ to the set of all families

$$(\mathcal{L}_{\mathbf{e}} \subset \mathcal{L}_{\mathbf{e}} \otimes_{\mathbb{C}} \mathbb{C}(\mathbf{e}))_{\mathbf{e} \in [\mathbf{e}]}$$

such that

(LM1) for all $\mathbf{e}$, $\mathcal{L}_{\mathbf{e}}$ is an $\mathbb{C} \otimes_{\mathbb{C}} \mathbb{C}(\mathbf{e})$ -submodule of $\mathcal{L}_{\mathbf{e}} \otimes_{\mathbb{C}} \mathbb{C}(\mathbf{e})$, and an $\mathbb{C}(\mathbf{e})$ -direct summand of rank $\mathbf{e}$;

(LM2) for all $\mathbf{e} < \mathbf{e}_i$, the natural arrow $\mathcal{L}_{\mathbf{e}} \otimes_{\mathbb{C}} \mathbb{C}(\mathbf{e}) \rightarrow \mathcal{L}_{\mathbf{e}_i} \otimes_{\mathbb{C}} \mathbb{C}(\mathbf{e}_i)$ induced by the inclusion $\mathcal{L}_{\mathbf{e}} \hookrightarrow \mathcal{L}_{\mathbf{e}_i}$, maps $\mathcal{L}_{\mathbf{e}}$ into $\mathcal{L}_{\mathbf{e}_i}$;

- (LM3) for all $\mathfrak{p}$, the isomorphism $\mathfrak{p} \otimes \mathfrak{p} \xrightarrow{\sim} \mathfrak{p}$ identifies $\mathfrak{p}$ with $\mathfrak{p}$;
 (LM4) for all $\mathfrak{p}$, the perfect $\mathfrak{p}$ -bilinear pairing

$$(\mathfrak{p} \otimes \mathfrak{p}) \times (\mathfrak{p} \otimes \mathfrak{p}) \xrightarrow{(\cdot, \cdot)} \mathfrak{p}$$

identifies $\mathfrak{p}^\perp$ with $\mathfrak{p}$ inside $\mathfrak{p} \otimes \mathfrak{p}$, here $\mathfrak{p}^\perp$ is the orthogonal complement of $\mathfrak{p}$ with respect to the above perfect pairing.

- (LM5) (Kottwitz condition) for all $\mathfrak{p}$, the element $\mathfrak{p} \in \mathfrak{p} \otimes \mathfrak{p}$ acts on $\mathfrak{p}$ as an $\mathfrak{p}$ -linear endomorphism with characteristic polynomial

$$\chi(\mathfrak{p} \cdot \mathfrak{p} - \mathfrak{p} | \mathfrak{p}) = (\mathfrak{p} + \mathfrak{p})(\mathfrak{p} - \mathfrak{p}) \in \mathfrak{p}[\mathfrak{p}].$$

- (ii) We have a natural embedding of the generic fiber into the naive local model:

$$\mathfrak{p}[\mathfrak{p}] \hookrightarrow \mathfrak{p}[\mathfrak{p}]. \quad (2.2.1)$$

We define the local model $\mathfrak{p}[\mathfrak{p}](\mathfrak{p}, \mathfrak{p})$ to be the scheme-theoretic closure of this embedding.

The following theorem is due to [Pap00, Arz09, Smi15, Yu19, Luo25], among others.

Theorem 2.2.2. *Let $(\mathfrak{p}, \mathfrak{p}) = (\mathfrak{p} - \mathfrak{p}, \mathfrak{p})$, and assume that $\mathfrak{p}$ is not $\mathfrak{p}$ -modular in (i)-(v) below.*

- (i) *The local model is flat, normal and Cohen-Macaulay of dimension $\mathfrak{p}$.*
- (ii) *The special fiber is a reduced scheme whose irreducible components are all normal and Cohen-Macaulay of dimension $\mathfrak{p} - \mathfrak{p}$.*
- (iii) *The point on the special fiber corresponding to $(\mathfrak{p} \otimes \mathfrak{p} \subset \mathfrak{p} \otimes \mathfrak{p})$ is referred to as the worst point. All irreducible components are smooth except at this point.*
- (iv) *The local model admits a moduli interpretation via the strengthened spin condition (see [Luo25, §1.4] for more detailed discuss in each case.).*
- (v) *The geometry of the special fiber varies according to the value of $\mathfrak{p}$:*
 - When $\mathfrak{p} = \mathfrak{p}$, the special fiber is irreducible.
 - When $\mathfrak{p} = \mathfrak{p} + \mathfrak{p}$ and $\mathfrak{p} = \mathfrak{p}$, the special fiber is smooth and irreducible,
 - When $\mathfrak{p} = \mathfrak{p}$ and $\mathfrak{p} = \mathfrak{p} - \mathfrak{p}$, the special fiber has three irreducible components.
 - When $\mathfrak{p}$ is strongly non-special, the special fiber has two irreducible components.
- (vi) *If $\mathfrak{p}$ is $\mathfrak{p}$ -modular, then the local model is smooth, connected, of dimension $\mathfrak{p}$, and does not contain the worst point.*

Proof. Parts (i)-(iv) are summarized in [Luo25, Theorem 1.3.2]. Part (v) is due to [Pap00] when $\mathfrak{p} = \mathfrak{p}$, [Arz09] when $\mathfrak{p} = \mathfrak{p} + \mathfrak{p}$ and $\mathfrak{p} = \mathfrak{p}$, [Yu19] when $\mathfrak{p} = \mathfrak{p}$ and $\mathfrak{p} = \mathfrak{p} - \mathfrak{p}$, [Luo25] in the remaining cases. See also [Luo25, Rem. 4.3.6] for a detailed discussion. Part (vi) is due to [PR09]. $\square$

2.3. Splitting model. We retain the assumptions in §2.2.

Definition 2.3.1. (i) We define the *Kr mer model* $\mathfrak{p}[\mathfrak{p}](\mathfrak{p}, \mathfrak{p})$ as the moduli functor that assigns to each $\mathfrak{p}$ -algebra $\mathfrak{p}$ the families

$$(\mathfrak{p} \subset \mathfrak{p} \subset \mathfrak{p} \otimes \mathfrak{p})_{\mathfrak{p} \in [\mathfrak{p}]}$$

such that

- (SP1) The filtration $(\mathfrak{p} \subset \mathfrak{p} \otimes \mathfrak{p})_{\mathfrak{p} \in [\mathfrak{p}]}$ satisfies (LM1)-(LM4);
- (SP2) $\mathfrak{p}$ is an $\mathfrak{p} \otimes \mathfrak{p}$ -submodule of $\mathfrak{p} \otimes \mathfrak{p}$ and an $\mathfrak{p}$ -direct summand of rank $\mathfrak{p}$;

(SP3) For each $\text{ant} < \text{bee}$, the morphism $\text{ant} \otimes \text{house} \otimes \text{earth} \rightarrow \text{ant} \otimes \text{house} \otimes \text{earth}$ maps $\text{ant} \otimes \text{house}$ into $\text{ant} \otimes \text{house}$:

$$\begin{array}{ccc} \text{ant} \otimes \text{house} \otimes \text{earth} & \longrightarrow & \text{ant} \otimes \text{house} \otimes \text{earth} \\ \uparrow & & \uparrow \\ \text{ant} \otimes \text{house} & \longrightarrow & \text{ant} \otimes \text{house} \end{array}$$

(SP4) The isomorphism $\text{ant} \otimes \text{house} \otimes \text{earth} \rightarrow \text{ant} \otimes \text{house} \otimes \text{earth}$ induced by $\text{ant} \otimes \text{house} \rightarrow \text{ant} \otimes \text{house}$ identifies $\text{ant} \otimes \text{house}$ with $\text{ant} \otimes \text{house}$;

(SP5) $(\text{ant} + \text{bee})(\text{ant} \otimes \text{house}) \subset \text{ant} \otimes \text{house}$;

(SP6) $(\text{ant} - \text{bee})(\text{ant} \otimes \text{house}) = (\text{ant})$.

(ii) We define the *naive splitting model* $\text{naive_splitting_model}(\text{ant}, \text{bee})$ as the closed subfunctor of $\text{naive_splitting_model}(\text{ant}, \text{bee})$ which associates to each house -algebra earth the families

$$(\text{ant} \otimes \text{house} \subset \text{ant} \otimes \text{house} \subset \text{ant} \otimes \text{house} \otimes \text{earth})_{\text{ant} \in [\text{ant}]} \in \text{naive_splitting_model}(\text{ant}, \text{bee})$$

which further satisfy:

(SP1') The filtration $(\text{ant} \otimes \text{house} \subset \text{ant} \otimes \text{house} \otimes \text{earth})_{\text{ant} \in [\text{ant}]}$ defines an earth -point of $\text{naive_splitting_model}(\text{ant}, \text{bee})$.

By Theorem 2.2.2(iv), this is equivalent to require that the filtration satisfies the strengthened spin condition, see [Luo25, §2].

(iii) We have a natural embedding of the generic fiber into the naive splitting model:

$$\text{naive_splitting_model}(\text{ant}, \text{bee}) \hookrightarrow \text{naive_splitting_model}(\text{ant}, \text{bee}).$$

We define the *splitting model* $\text{splitting_model}(\text{ant}, \text{bee})$ to be the scheme-theoretic closure of this embedding.

Remark 2.3.2. The space $\text{splitting_model}(\text{ant}, \text{bee})$ was initially studied by [Krä03] when $\text{bee} = \text{ant}$ and $(\text{ant}, \text{bee}) = (\text{ant} - \text{bee}, \text{bee})$. When $\text{bee} = \text{ant}$, $\text{splitting_model}(\text{ant}, \text{bee})$ is flat. This is proved in loc.cit. for $(\text{ant}, \text{bee}) = (\text{ant} - \text{bee}, \text{bee})$, and extended to all (ant, bee) in [BH25].

For $(\text{ant}, \text{bee}) = (\text{ant} - \text{bee}, \text{bee})$ and strongly non-special bee , we will show that $\text{splitting_model}(\text{ant} - \text{bee}, \text{bee}) \simeq \text{splitting_model}(\text{ant} - \text{bee}, \text{bee})$, see Theorem 3.3.4. In general, there is a chain of closed embeddings

$$\text{splitting_model}(\text{ant}, \text{bee}) \hookrightarrow \text{splitting_model}(\text{ant}, \text{bee}) \hookrightarrow \text{splitting_model}(\text{ant}, \text{bee}),$$

each of which becomes an isomorphism over the generic fiber. Therefore, $\text{splitting_model}(\text{ant}, \text{bee})$ can also be defined as the scheme-theoretic closure of the open embedding:

$$\text{splitting_model}(\text{ant}, \text{bee}) \hookrightarrow \text{splitting_model}(\text{ant}, \text{bee}).$$

It is worth noting that for general (ant, bee) and bee , the existence of the forgetful functor for $\text{splitting_model}(\text{ant}, \text{bee})$ as in (2.4.1) is unclear, though such a functor would be useful in practice.

2.4. Basic properties. We have the forgetful functor

$$\text{forgetful} : \text{splitting_model}(\text{ant}, \text{bee}) \rightarrow \text{splitting_model}(\text{ant}, \text{bee}) \otimes \text{house}, \quad (\text{ant} \otimes \text{house}, \text{ant} \otimes \text{house}) \mapsto \text{ant} \otimes \text{house}. \quad (2.4.1)$$

Lemma 2.4.1. forgetful induces an isomorphism over the generic fibers.

Proof. Since $\text{house} \otimes \text{house} \simeq \text{house} \times \text{house}$ is split, when earth is a house -algebra, the additional house -action induces a house -grading on $\text{ant} \otimes \text{house}$ and $\text{ant} \otimes \text{house}$. This grading is the same as the additional filtration in the naive splitting model. Moreover, the degree bee part of the grading can be recovered from the degree ant part by taking dual. Hence both the generic fiber of the canonical local model and the naive splitting model are isomorphic to the Grassmannian $\text{Gr}(\text{ant}, \text{bee})$. $\square$

2.4.2. Let $\mathbb{E}$ be an $\mathcal{H}$ -algebra and $\mathcal{H}$ a locally free $\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E}$ -module of rank r . We have an exact sequence

$$\mathcal{H} \rightarrow (\mathcal{H} - \mathcal{H})_{\mathcal{H}} \rightarrow \mathcal{H} \xrightarrow{+} (\mathcal{H} + \mathcal{H})_{\mathcal{H}} \rightarrow \mathcal{H}.$$

This induces an isomorphism:

$$(\mathcal{H} + \mathcal{H})_{\mathcal{H}} \xrightarrow{\sim} \mathcal{H}/(\mathcal{H} - \mathcal{H})_{\mathcal{H}}, \quad \mathcal{H} \mapsto (\mathcal{H} + \mathcal{H})^{-1} \mathcal{H} + (\mathcal{H} - \mathcal{H})_{\mathcal{H}}.$$

See [How19, Lemma 3.2]. The same result holds if we change $\mathcal{H} \pm \mathcal{H}$ to $\mathcal{H} \mp \mathcal{H}$. In particular, $(\mathcal{H} - \mathcal{H})_{\mathcal{H}}$ and $(\mathcal{H} + \mathcal{H})_{\mathcal{H}}$ are both $\mathbb{E}$ -direct summands of rank r . Note that these two submodules specialize to the submodule $\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} \subset \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E}$ over the special fiber.

Now we come back to the perfect pairing for any $\mathcal{H}$ -algebra $\mathbb{E}$:

$$\langle, \rangle : \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} \times \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} \rightarrow \mathbb{E}.$$

Since

$$\langle \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E}, \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} \rangle = \langle \mathcal{H}, - \mathcal{H} \rangle, \quad \forall \mathcal{H} \in \mathcal{H}, \mathcal{H} \in \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E}. \quad (2.4.2)$$

We have

$$\left((\mathcal{H} \pm \mathcal{H})_{\mathcal{H}} \otimes_{\mathcal{H}} \mathbb{E} \right)^{\perp} = (\mathcal{H} \pm \mathcal{H})_{\mathcal{H}} \otimes_{\mathcal{H}} \mathbb{E}.$$

We define an induced pairing,

$$\{, \} : (\mathcal{H} - \mathcal{H})_{\mathcal{H}} \otimes_{\mathcal{H}} \mathbb{E} \times (\mathcal{H} + \mathcal{H})_{\mathcal{H}} \otimes_{\mathcal{H}} \mathbb{E} \rightarrow \mathbb{E}, \quad \{ \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E}, \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} \} \mapsto \langle \mathcal{H}, (\mathcal{H} + \mathcal{H})^{-1} \mathcal{H} \rangle. \quad (2.4.3)$$

For any $\mathbb{E}$ -point $(\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} \subset \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E})_{\mathcal{H} \in [\mathcal{H}]}$ of $\mathcal{H}[\mathcal{H}]$, Axiom (SP6) implies that $\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} \subset (\mathcal{H} + \mathcal{H})_{\mathcal{H}} \otimes_{\mathcal{H}} \mathbb{E}$. We define

$$\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} := \left((\mathcal{H} + \mathcal{H})^{-1} \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} \right)^{\perp} \subset \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E}. \quad (2.4.4)$$

The following proposition follows from the proof of [BH25, Proposition 2.4], which is also implicitly proved in [How19, §3].

Proposition 2.4.3. (i) The submodule $\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E}$ is a locally free $\mathbb{E}$ -module of rank r and $\mathbb{E}$ -direct summand of $\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E}$. Moreover, we have

$$(\mathcal{H} - \mathcal{H})_{\mathcal{H}} \otimes_{\mathcal{H}} \mathbb{E} \subseteq \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E}, \quad (\mathcal{H} + \mathcal{H})_{\mathcal{H}} \otimes_{\mathcal{H}} \mathbb{E} = (\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E}).$$

(ii) Let $\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} \subset \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} / \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E}$ be the isotropic complement of $\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} \subset \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E}$ with respect to the perfect pairing

$$\langle, \rangle : \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} \times (\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} / \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E}) \rightarrow \mathbb{E}.$$

Then $\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E}$ is a direct $\mathbb{E}$ -summand of rank r , with

$$(\mathcal{H} - \mathcal{H})_{\mathcal{H}} \otimes_{\mathcal{H}} \mathbb{E} / \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} \subset \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E}, \quad (\mathcal{H} + \mathcal{H})_{\mathcal{H}} \otimes_{\mathcal{H}} \mathbb{E} = (\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E}).$$

Furthermore, we can characterize $\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E}$ using the following formula:

$$\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} = \left(\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} / \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} \xrightarrow{+} (\mathcal{H} - \mathcal{H})_{\mathcal{H}} \otimes_{\mathcal{H}} \mathbb{E} / \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} \right)$$

Proof. (i) It follows from the same proof as [BH25, Proposition 2.4], note that the rank argument follows from (2.4.3).

(ii) It is clear from (i) and (2.4.2) that we have the inclusion

$$\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} \subset \left(\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} / \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} \xrightarrow{+} (\mathcal{H} - \mathcal{H})_{\mathcal{H}} \otimes_{\mathcal{H}} \mathbb{E} / \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} \right)$$

By (i) we know that $\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E}$ is an $\mathbb{E}$ -direct summand of rank r . Since $\mathcal{H} \otimes_{\mathcal{H}} \mathbb{E} \xrightarrow{+} (\mathcal{H} - \mathcal{H})_{\mathcal{H}} \otimes_{\mathcal{H}} \mathbb{E} / \mathcal{H} \otimes_{\mathcal{H}} \mathbb{E}$ is a surjection, the kernel is also an $\mathbb{E}$ -direct summand of rank r , hence we get the equality. $\square$

It is clear that the morphism $\text{🏠} \otimes \text{🏠} \rightarrow \text{🏠} \otimes \text{🏠}$ maps 🌍 into 🌍 for any $\text{🐜} \in [\text{🐜}]$ such that $\text{🐜} < \text{🐜}$. Denote by $\overline{(-)}$ the conjugate on 🌍 , we have the following morphism

$$\text{🏠}(\text{🌸}, \text{🌿}) \xrightarrow{\sim} \text{🏠}(\text{🌿}, \text{🌸}), \quad (\text{🌍} \subset \text{🌍})_{\text{🐜} \in [\text{🐜}]} \mapsto (\overline{\text{🌍}} \subset \overline{\text{🌍}})_{\text{🐜} \in [\text{🐜}]}.$$

Note that we conjugated the 🌍 -action on the additional filtration $\overline{\text{🌍}}$ so that the Krämer conditions (SP5) and (SP6) are satisfied. By reversing this procedure, we see that the morphism is an isomorphism. This isomorphism restricts to an isomorphism

$$\text{🏠}(\text{🌸}, \text{🌿}) \xrightarrow{\sim} \text{🏠}(\text{🌿}, \text{🌸}).$$

Remark 2.4.4. Similar to [BH25] and [BZZ25], we can define locally closed subschemes

$$\text{🏠}_{\text{🐜}} := \{ \text{🌸} \in \text{🏠}(\text{🌸}, \text{🌿}) \mid \text{🌍} \cap \text{🌍} = \text{🌍}, \text{🌍} \cap \text{🌍} = \text{🌍} \}, \quad \text{🐜} \in [\text{🐜}].$$

One may ask the following questions:

- (i) What is the relation between $\text{🏠}_{\text{🐜}}$ for different 🐜 ?
- (ii) For every $\text{🐜} \leq \text{🐜}$, we have the inclusion

$$\overline{\text{🏠}_{\text{🐜}}} \subseteq \bigcup_{\text{🐜} \leq \text{🐜}' \leq \text{🐜} \leq \text{🐜}'} \text{🏠}_{\text{🐜}'}$$

It this an equality?

- (iii) Let $\text{🏠}_{\text{🐜}} := \text{🏠}_{\text{🐜}} \cap \text{🏠}(\text{🌸}, \text{🌿})$. Is $\text{🏠}_{\text{🐜}}$ smooth? What is the dimension of $\text{🏠}_{\text{🐜}}$?
- (iv) How do these indices relate to the Kottwitz–Rapoport strata?

2.5. The case of signature $(\text{🐜} - \text{🌿}, \text{🌿})$. From now on, we assume $\text{🐜} \geq \text{🌿}$ and $(\text{🌸}, \text{🌿}) = (\text{🐜} - \text{🌿}, \text{🌿})$, and omit the signature when writing $\text{🏠}_{\text{🐜}}$, $\text{🏠}_{\text{🐜}}$ and $\text{🏠}_{\text{🐜}}$. In this case, we have $\text{🌍} = \text{🌍}$, hence the canonical local model and the splitting model are defined in the same base ring.

Suppose 🐜 is not 🌿 -modular, let $\text{🌍} \in \text{🏠}_{\text{🐜}}(\text{🌸}, \text{🌿})$ be the worst point defined as $\{ \text{🌍} \in \text{🏠}_{\text{🐜}}(\text{🌸}, \text{🌿}) \mid \text{🌍} \cap \text{🌍} = \text{🌍} \}$.

Proposition 2.5.1. Suppose 🐜 is not 🌿 -modular, the map $\text{🌍} \rightarrow \text{🌍}$ induces an isomorphism

$$\text{🌍} : \text{🏠}_{\text{🐜}} \setminus \text{🌍} \rightarrow \text{🏠}_{\text{🐜}} \setminus \{ \text{🌍} \}.$$

Proof. For each $\text{🐜} \in [\text{🐜}]$, we have a sheaf of module $(\text{🌍} + \text{🌍})(\text{🌍})$ over $\text{🏠}_{\text{🐜}}$. By (SP5), we have $(\text{🌍} + \text{🌍})(\text{🌍}) \subseteq \text{🌍}$. We claim that a point $\text{🐜} \in \text{🏠}_{\text{🐜}} \setminus \{ \text{🌍} \}$ corresponds to a $\text{🐜}(\text{🐜})$ -subspace of $\text{🌍} \otimes \text{🌍}$ such that $(\text{🌍} + \text{🌍})(\text{🌍}) \neq (\text{🌍})$:

- (i) If 🐜 lies in the generic fiber, the subspace $(\text{🌍} + \text{🌍})(\text{🌍})$ is locally free of rank 🌿 by the Kottwitz condition.
- (ii) If 🐜 lies in the special fiber, the claim follows from the definition of the worst point.

The remaining proof is the same as that of [Krä03, Proposition 4.3]. $\square$

Definition 2.5.2. We define the *naive exceptional divisor* as $\text{🌍} := \text{🌍}^{-1}(\text{🌍}) \subset \text{🏠}_{\text{🐜}}$, and the *exceptional divisor* as the scheme-theoretic intersection $\text{🌍} := \text{🌍}^{-1}(\text{🌍}) \cap \text{🏠}_{\text{🐜}}$.

Remark 2.5.3. When 🐜 is 🌿 -modular, as the canonical local model does not contain the worst point, the same argument as in Proposition 2.5.1 implies that the splitting model is isomorphic to the local model, and the exceptional divisors are empty.

Proposition 2.5.4. Suppose $\text{🐜} \notin \{ \text{🌿}, \text{🐜} \}$. Let 🌍 and $\text{🌍}_{\text{🐜}}$ be the transition maps $\text{🏠}_{\text{🐜}} \rightarrow \text{🏠}_{\text{🐜}}$ and $\text{🏠}_{\text{🐜}} \rightarrow \text{🏠}_{\text{🐜}}$ respectively. The naive exceptional divisor is a linked variety with equal dimension $\text{🐜} - \text{🌿}$. It has two irreducible components $\text{🌍}_{\text{🐜}}$ and $\text{🌍}_{\text{🐜}}$, defined by $\text{🌍}_{\text{🐜}}(\text{🌍}) = (\text{🌍})$ and $\text{🌍}_{\text{🐜}}(\text{🌍}) = (\text{🌍})$ respectively. Their intersection $\text{🌍}_{\text{🐜}} \cap \text{🌍}_{\text{🐜}}$ is irreducible of dimension $\text{🐜} - \text{🌿}$.

Proof. The transition maps induce the transition maps between the lattices which represent the worst point:

$$\text{bell} : \text{house} \rightarrow \text{house} - \text{key}, \quad \text{bell} - \text{key} : \text{house} \rightarrow \text{house} + \text{key}.$$

By choosing the standard basis as in [Luo25, §3.1], the transition maps can be represented by matrices

$$\text{bell} = \begin{pmatrix} \text{cube} & & \\ & \text{egg} & \\ & & \text{cube} - \text{key} \end{pmatrix}, \quad \text{bell} - \text{key} = \begin{pmatrix} \text{egg} & & \\ & \text{cube} - \text{key} & \\ & & \text{egg} \end{pmatrix}. \quad (2.5.1)$$

From the definition, the naive exceptional divisor has the moduli description which associates to each cube -algebra egg the families

$$(\text{egg}, \text{egg}) \in \text{house}(\text{house} - \text{key}) \times \text{house}(\text{house} - \text{key})$$

such that $\text{bell}(\text{egg}) \subset \text{egg}$ and $\text{bell} - \text{key}(\text{egg}) \subset \text{egg}$ under the identification $\text{house} \xrightarrow{\sim} \text{house}$ of (LM3) and (SP4). All the other axioms hold automatically. The statements now is standard. For instance, the naive exceptional locus agrees with the special fiber of the local model for house with cocharacter $\text{key} = (\text{key}, \text{egg} - \text{key})$ and lattice chain generated by $\text{house} - \text{key}$ and $\text{house} + \text{key}$, see [Gör01]. $\square$

Remark 2.5.5. (i) When $\text{cube} = \{\text{egg}\}$, the native splitting model is identical to the splitting model, this is studied in [Krä03].

(ii) When $\text{key} = \text{key}$ and $\text{cube} = \{\text{key}\}$, the canonical local model is isomorphic to the (naive) splitting model, since it doesn't contain the worst point.

(iii) When $\text{key} = \text{key} + \text{key}$ and $\text{cube} = \{\text{key}\}$, the canonical local model is smooth. However, canonical local model, splitting model, and naive splitting model, are not isomorphic to each other. A direct calculation, or using Theorem 4.0.1 and [Ric13, §5.2], one can get a moduli interpretation in this case.

(iv) When $\text{key} = \text{key}$ and $\text{cube} = \{\text{key} - \text{key}\}$, [RSZ18, Proposition 9.12] establishes the isomorphism:

$$\text{house}[\text{key} - \text{key}] \simeq \text{house}[\text{key} - \text{key}, \text{key}].$$

Yu's thesis [Yu19] introduces the moduli space $\text{house}[\text{key} - \text{key}, \text{key}]$ over $\text{house}[\text{key} - \text{key}, \text{key}]$ parameterizing additional filtrations $(\text{egg} \subset \text{egg})_{\text{key} \in [\text{key} - \text{key}, \text{key}]}$ that satisfy the Krämer condition and, for any $\text{key}, \text{key} \in [\text{key} - \text{key}, \text{key}]$ the transition maps

$$\text{bell} : \text{house} \otimes \text{house} \rightarrow \text{house} \otimes \text{house}$$

carry $\text{bell}(\text{egg}) \subseteq \text{egg}$. Yu proves that $\text{house}[\text{key} - \text{key}, \text{key}]$ is flat over house . In this setting, a similar isomorphism holds in the splitting model context:

$$\text{house}[\text{key} - \text{key}] \simeq \text{house}[\text{key} - \text{key}, \text{key}]. \quad (2.5.2)$$

In fact, Proposition 2.5.1 and its proof shows that the forgetful map

$$\text{house}[\text{key} - \text{key}, \text{key}] \rightarrow \text{house}[\text{key} - \text{key}]$$

is a closed embedding, as we always have $\text{egg} = (\text{key} + \text{key})\text{egg}$. This map induces an isomorphism over the generic fiber, and the isomorphism (2.5.2) follows by taking the flat closure.

It is noteworthy that while the additional filtration $\text{egg} \subset \text{egg}$ always exists in $\text{house}[\text{key} - \text{key}, \text{key}]$, it may not necessarily satisfy:

$$\text{bell} - \text{key}(\text{egg}) \subseteq \text{egg}, \quad \text{bell}(\text{egg}) \subseteq \text{egg} + \text{key}.$$

Yu applies these conditions to $\text{house}[\text{key} - \text{key}]$, and get the closed subscheme $\text{house}[\text{key} - \text{key}] = \text{house}[\text{key} - \text{key}, \text{key}]$.

2.5.6. Consider the base change of $\mathcal{C}_{[m],z}$ to the special fiber

$$\mathcal{C}_{[m],z} = \mathcal{C}_{[m],z} : \mathcal{C}_{[m],z} \longrightarrow \mathcal{C}_{[m],z}.$$

Denote $\mathcal{C}_{[m],z} = \bigcup_{\mathcal{C}_{[m],z}} \mathcal{C}_{[m],z}$ as the union of irreducible components. Define $\widetilde{\mathcal{C}_{[m],z}}$ as the scheme-theoretic closure of $\mathcal{C}_{[m],z} \setminus \{\mathcal{C}_{[m],z}\}$ for each $\mathcal{C}_{[m],z}$. Then we have

$$\mathcal{C}_{[m],z}(\widetilde{\mathcal{C}_{[m],z}}) = \mathcal{C}_{[m],z} \quad \text{for any } \mathcal{C}_{[m],z}.$$

Proposition 2.5.7. *The special fiber $\mathcal{C}_{[m],z}$ of the naive splitting model is the union of $\widetilde{\mathcal{C}_{[m],z}}$ and exceptional divisors:*

$$\mathcal{C}_{[m],z} = \mathcal{C}_{[m],z} \cup \mathcal{C}_{[m],z} \cup \bigcup_{\mathcal{C}_{[m],z}} \widetilde{\mathcal{C}_{[m],z}}.$$

Proof. It is clear from the construction that $\mathcal{C}_{[m],z}$ is the union of these closed subschemes: for any $\mathcal{C}_{[m],z}$ -point $\mathcal{C}_{[m],z} \in \mathcal{C}_{[m],z}(\mathcal{C}_{[m],z})$, if $\mathcal{C}_{[m],z} \in \mathcal{C}_{[m],z} \setminus \{\mathcal{C}_{[m],z}\}$, then $\mathcal{C}_{[m],z} \in \bigcup_{\mathcal{C}_{[m],z}} \widetilde{\mathcal{C}_{[m],z}}$, otherwise, $\mathcal{C}_{[m],z} \in \mathcal{C}_{[m],z} = \mathcal{C}_{[m],z} \cup \mathcal{C}_{[m],z}$.

Next, since $\mathcal{C}_{[m],z} \setminus \{\mathcal{C}_{[m],z}\}$ is irreducible of dimension $\mathcal{C}_{[m],z} - \mathcal{C}_{[m],z}$ for each $\mathcal{C}_{[m],z}$, so is $\widetilde{\mathcal{C}_{[m],z}}$ for each $\mathcal{C}_{[m],z}$. We have shown that $\mathcal{C}_{[m],z}$ is irreducible of dimension $\mathcal{C}_{[m],z} - \mathcal{C}_{[m],z}$ for $\mathcal{C}_{[m],z} = \mathcal{C}_{[m],z}, \mathcal{C}_{[m],z}$ in Proposition 2.5.4. It is clear that they are irreducible components. $\square$

3. REGULARITY OF THE SPLITTING MODEL

3.1. Recollection of the canonical local model. From now on, we will always assume the signature to be $(\mathcal{C}_{[m],z} - \mathcal{C}_{[m],z}, \mathcal{C}_{[m],z})$, and omit the signature when writing $\mathcal{C}_{[m],z}, \mathcal{C}_{[m],z}, \mathcal{C}_{[m],z}$ and $\mathcal{C}_{[m],z}$. We assume the index $\mathcal{C}_{[m],z} = \{\mathcal{C}_{[m],z}\}$ is not $\mathcal{C}_{[m],z}$ -modular.

We follow the discussion in [Luo25, §3]. We will choose the *standard* ordered $\mathcal{C}_{[m],z}$ -basis as in [Luo25, (3.1.1)]:

$$\begin{aligned} \mathcal{C}_{[m],z} : & \mathcal{C}_{[m],z}^{-1} \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z}^{-1} \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}; \\ & \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}. \\ \mathcal{C}_{[m],z} : & \mathcal{C}_{[m],z}^{-1} \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z}^{-1} \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}; \\ & \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}. \end{aligned} \tag{3.1.1}$$

We will also choose the *reordered* basis as in [Luo25, (3.1.3)]:

$$\begin{aligned} \mathcal{C}_{[m],z} : & \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \mathcal{C}_{[m],z}^{-1} \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z}^{-1} \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}; \\ & \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \\ & \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}; \\ & \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}. \\ \mathcal{C}_{[m],z} : & \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \mathcal{C}_{[m],z}^{-1} \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z}^{-1} \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}; \\ & \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \\ & \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}; \\ & \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \dots, \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}. \end{aligned} \tag{3.1.2}$$

The *worst point* $\mathcal{C}_{[m],z} \in \mathcal{C}_{[m],z}(\mathcal{C}_{[m],z})$ is defined as

$$\mathcal{C}_{[m],z} = \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z} \subset \mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z}, \quad \text{for any } \mathcal{C}_{[m],z} \in [\mathcal{C}_{[m],z}],$$

see [Luo25, Lemma 3.1.1]. The naive local model embeds into $\mathcal{C}_{[m],z}(\mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z})$, and we choose an open affine chart $\mathcal{C}_{[m],z} \subset \mathcal{C}_{[m],z}(\mathcal{C}_{[m],z} \otimes \mathcal{C}_{[m],z})$ of the worst point. In the chart, the $\mathcal{C}_{[m],z}$ and $\mathcal{C}_{[m],z}$ are represented by the $\mathcal{C}_{[m],z} \times \mathcal{C}_{[m],z}$ matrices $\begin{pmatrix} \mathcal{C}_{[m],z} \\ \mathcal{C}_{[m],z} \end{pmatrix}$ and $\begin{pmatrix} \mathcal{C}_{[m],z} \\ \mathcal{C}_{[m],z} \end{pmatrix}$ with respect to the reordered basis of

🏠🔑 and $\text{🏠🔑} - \text{🏠🔑}$ respectively. The worst point corresponds to $\text{🏠} = \text{🍌}$. We define 🌂🔑 (resp. 🌂🔑) as the scheme-theoretic intersection $\text{🌂🔑} \cap \text{🏠🔑}(\text{🐛} - \text{🌱}, \text{🌱})$ (resp. $\text{🌂🔑} \cap \text{🏠🔑}(\text{🐛} - \text{🌱}, \text{🌱})$). We further give the partition of the matrix

$$\text{🏠} = \left(\begin{array}{c|c} \text{🏠🔑} & \text{🏠🔑} \\ \hline \text{🏠🔑} & \text{🏠🔑} \end{array} \right) = \left(\begin{array}{c|c} \text{🚀🔑} & \text{🌸🔑} \\ \hline \text{🌴🔑} & \text{🏠🔑} \end{array} \right).$$

We will also denote 🌸 and 🌸 as the corresponding matrices with respect to the standard basis. By Axiom (LM4), we have:

$$\text{🌸} = \left(\begin{array}{c|c} \text{🚀} & \text{🏠} \\ \hline \text{🌴} & \text{🏠} \end{array} \right), \quad \text{🌸} = \text{🐛} \text{🌸} - \text{🌱} \text{🌸} = \left(\begin{array}{c|c} \text{🚀} & \text{🌸} \\ \hline \text{🌴} & \text{🏠} \end{array} \right), \quad (3.1.3)$$

where 🐛 defined in §1.9 is the matrix representing the symmetric form. For later use, we denote

$$\left(\begin{array}{c} \text{🌸} \\ \text{🌸} \end{array} \right) := (\text{🐛}, \dots, \text{🐛}), \quad \left(\begin{array}{c} \text{🌸} \\ \text{🌸} \end{array} \right) := (\text{🐛}, \dots, \text{🐛}). \quad (3.1.4)$$

Proposition 3.1.1. ([Luo25, Theorem 8.0.1]) Assume 🔑 is not 🍌 -modular, the affine ring of 🌂🔑 is isomorphic to the factor ring of 🏠🔑 modulo the ideal generated by the entries of the following matrices:

- (i) $\text{🌸} = \text{🍌}$;
- (ii) $\text{🏠🔑} + \text{🏠🔑} = \text{🍌}$, $-\text{🏠🔑} + \text{🏠🔑} = \text{🍌}$, $\text{🏠🔑} + \text{🏠🔑} = \text{🍌}$, $\text{🏠🔑} - \text{🏠🔑} = \text{🍌}$;
- (iii) $\text{🏠🔑} - \text{🏠🔑} = \text{🍌}$, $\text{🏠🔑} - \text{🏠🔑} = \text{🍌}$, $\text{🏠🔑} + \text{🏠🔑} = \text{🍌}$, $\text{🏠🔑} - \text{🏠🔑} = \text{🍌}$;
- (iv) $\text{🌂🔑} + \text{🍌} = \text{🍌}$, $\text{🌂🔑} - \text{🍌} = \text{🍌}$;
- (v) $\text{🌂} = \text{🌂}$, $\text{🌂} = \text{🌂}$, $\text{🌂} = -\text{🌂} - \text{🌂} = \text{🌂}$, $\text{🌂} = -\text{🌂}$, $\text{🌂} = \text{🌂}$, $\text{🌂} = -(\text{🐛} - \text{🌱} - \text{🌱})$.

Here (i)-(iii) follows from (LM1)-(LM4), while (iv)-(v) follows from the strengthened spin condition [Luo25]. $\square$

Remark 3.1.2. When 🔑 is strongly non-special, by Theorem 2.2.2 (2), the special fiber has two irreducible components 🌱 and 🐛 , they are defined by $\text{🏠🔑} = \text{🍌}$ and $\text{🏠🔑} = \text{🍌}$, resp. For the other cases, by [Luo25, Remark 4.3.6], over the special fiber, we have:

- (i) When $\text{🔑} = \text{🍌}$, we have 🏠🔑 is of size $\text{🍌} \times \text{🍌}$, hence only 🏠🔑 left. Hence the special fiber is irreducible.
- (ii) When $\text{🐛} = \text{🌱} + \text{🌱}$ and $\text{🔑} = \text{🌱}$, we have $\text{🏠🔑} = \text{🍌}$ for free, and the special fiber is smooth and irreducible.
- (iii) When $\text{🐛} = \text{🌱}$ and $\text{🔑} = \text{🌱} - \text{🌱}$, the matrix 🏠🔑 is of the form

$$\text{🏠🔑} = \left(\begin{array}{c} \text{🌸} \\ \text{🌸} \end{array} \right).$$

In particular, $\text{🏠🔑} = \text{🍌}$ further splits into two irreducible components, which is Yu's case [Yu19].

3.2. Kramer model relations. Let 👉 be non- 👉 -modular. We define an open subscheme $\text{👉} \subset \text{👉}$ as the following pull-back:

$$\begin{array}{ccc} \text{👉} & \xrightarrow{\quad} & \text{👉} \\ \downarrow & \square & \downarrow \\ \text{👉} & \xrightarrow{\quad} & \text{👉}(\text{👉}, \text{👉} \otimes \text{👉}). \end{array}$$

The right-side map is defined by the composition:

$$\text{👉} \rightarrow \text{👉} \hookrightarrow \text{👉}(\text{👉}, \text{👉} \otimes \text{👉}).$$

We use an analogous pull-back diagram to define 👉 and 👉 . In this subsection, we write down the defining equations of 👉 .

We will use the standard basis of 👉 and 👉 in (3.1.1).

Let us consider an 👉 -point in 👉 that can be represented as $(\text{👉} \subset \text{👉}; \text{👉} \subset \text{👉})$. We assume 👉 and 👉 are respectively generated by two vectors whose coordinates with respect to the standard basis are

$$\text{👉} = (\text{👉} \cdots \text{👉})^T \in \text{👉}, \quad \text{👉} = (\text{👉} \cdots \text{👉})^T \in \text{👉}.$$

We can in addition assume that there exist indices $\text{👉}'$ and $\text{👉}'$ in the set $\{\text{👉}, \dots, \text{👉}\}$ such that $\text{👉}' = \text{👉}$ and $\text{👉}' = \text{👉}$.

3.2.1. Consider the condition: $\text{👉} \subset \text{👉}, \text{👉} \subset \text{👉}$. Using (3.1.4), we can find $\text{👉}, \text{👉} \in \text{👉}$ for $\text{👉}, \text{👉} = \text{👉}, \dots, \text{👉}$ such that

$$\text{👉} = \sum_{\text{👉}=\text{👉}} \text{👉} \text{👉}, \quad \text{👉} = \sum_{\text{👉}=\text{👉}} \text{👉} \text{👉}. \quad (3.2.1)$$

Denote $\text{👉} = (\text{👉} \cdots \text{👉})^T$ and $\text{👉} = (\text{👉} \cdots \text{👉})^T$. Then (3.2.1) becomes

$$\begin{pmatrix} \text{👉} \\ \text{👉} \end{pmatrix} \text{👉} = \text{👉}, \quad \begin{pmatrix} \text{👉} \\ \text{👉} \end{pmatrix} \text{👉} = \text{👉}. \quad (3.2.2)$$

One immediately deduces the following relations

$$\text{👉} = \text{👉} + \text{👉}, \dots, \text{👉} = \text{👉} + \text{👉} \quad \text{and} \quad \text{👉} = \text{👉} + \text{👉}, \dots, \text{👉} = \text{👉} + \text{👉}.$$

3.2.2. Next consider (SP6): $(\text{👉} - \text{👉})\text{👉} = (\text{👉}), (\text{👉} - \text{👉})\text{👉} = (\text{👉})$. By (LM1), we have $\text{👉} \subset \text{👉}$ ($\text{👉} = \text{👉}, \text{👉} - \text{👉}$). Since we choose the standard basis, we have

$$\text{👉} \begin{pmatrix} \text{👉} \\ \text{👉} \end{pmatrix} = \begin{pmatrix} \text{👉} \\ \text{👉} \end{pmatrix} \quad \text{and} \quad \text{👉} \begin{pmatrix} \text{👉} \\ \text{👉} \end{pmatrix} = \begin{pmatrix} \text{👉} \\ \text{👉} \end{pmatrix}.$$

Assuming (LM1), then (SP6) is equivalent to

$$\text{👉} \text{👉} = \text{👉}, \quad \text{👉} \text{👉} = \text{👉}. \quad (3.2.3)$$

Hence we conclude that $\text{👉} = \begin{pmatrix} \text{👉} \\ \text{👉} \end{pmatrix}, \text{👉} = \begin{pmatrix} \text{👉} \\ \text{👉} \end{pmatrix}$. The affine chart we are working on now can be assumed as $\text{👉} = \text{👉} = \text{👉}$ for some $\text{👉}, \text{👉} \in \{\text{👉}, \dots, \text{👉}\}$.

3.2.3. Next consider (SP5): $(\text{🍷} + \text{🍷})\text{🍷} \subset \text{🍷}, (\text{🍷} + \text{🍷})\text{🍷} \subset \text{🍷}$. This is equivalent to the existence of additional parameters $\text{🍷} = (\text{🍷} \cdots \text{🍷})^*$ and $\text{🍷} = (\text{🍷} \cdots \text{🍷})^*$ such that

$$\begin{pmatrix} \text{🍷} \\ \text{🍷} \end{pmatrix} + \text{🍷} \begin{pmatrix} \text{🍷} \\ \text{🍷} \end{pmatrix} = \begin{pmatrix} \text{🍷} \\ \text{🍷} \end{pmatrix} \cdot \text{🍷}, \quad \begin{pmatrix} \text{🍷} \\ \text{🍷} \end{pmatrix} + \text{🍷} \begin{pmatrix} \text{🍷} \\ \text{🍷} \end{pmatrix} = \begin{pmatrix} \text{🍷} \\ \text{🍷} \end{pmatrix} \cdot \text{🍷}.$$

Or equivalently

$$\text{🍷} + \text{🍷} = \text{🍷}, \quad \text{🍷} + \text{🍷} = \text{🍷}. \quad (3.2.4)$$

It is clear that (SP5) and (SP6) imply that 🍷 is 🍷 -stable for $\text{🍷} = \text{🍷} - \text{🍷}$. Now (3.2.3) and (3.2.4) together imply

$$\text{🍷} \text{🍷} = \text{🍷}, \quad \text{🍷} \text{🍷} = \text{🍷}.$$

Recall that we have assumed $\text{🍷} = \text{🍷}$ and $\text{🍷} = \text{🍷}$. By comparing the 🍷 -th row and the 🍷 -th row of $\text{🍷} \text{🍷} = \text{🍷}$ and $\text{🍷} \text{🍷} = \text{🍷}$, resp., we see that the above equations are equivalent to

$$\text{🍷} = \text{🍷}, \quad \text{🍷} = \text{🍷}. \quad (3.2.5)$$

Note that (3.2.4) and (3.2.5) imply (3.2.3).

Finally, denote

$$\text{🍷} = \begin{pmatrix} \text{🍷} \\ \text{🍷} \\ \text{🍷} \end{pmatrix}, \quad \text{🍷} = \begin{pmatrix} \text{🍷} \\ \text{🍷} \\ \text{🍷} \end{pmatrix}, \quad \text{🍷} = \begin{pmatrix} \text{🍷} \\ \text{🍷} \\ \text{🍷} \end{pmatrix}, \quad \text{🍷} = \begin{pmatrix} \text{🍷} \\ \text{🍷} \\ \text{🍷} \end{pmatrix}$$

where all parts are of size $\text{🍷} \times \text{🍷}$, $(\text{🍷} - \text{🍷}) \times \text{🍷}$ and $\text{🍷} \times \text{🍷}$ respectively. Then (SP3) is equivalent to:

$$\begin{pmatrix} \text{🍷} \\ \text{🍷} \\ \text{🍷} \end{pmatrix} \in \text{🍷} \begin{pmatrix} \text{🍷} \\ \text{🍷} \\ \text{🍷} \end{pmatrix}, \quad \begin{pmatrix} \text{🍷} \\ \text{🍷} \\ \text{🍷} \end{pmatrix} \in \text{🍷} \begin{pmatrix} \text{🍷} \\ \text{🍷} \\ \text{🍷} \end{pmatrix}. \quad (3.2.6)$$

3.3. Krämer condition implies the strengthened spin condition. In this subsection, we will show that closed embedding $\text{🍷} \hookrightarrow \text{🍷}$ is an isomorphism. Recall that in Definition 2.3.1, we only impose Axioms (LM1)-(LM4) to 🍷 and 🍷 in 🍷 . By Proposition 3.1.1, we only have the following relations for 🍷

- (i) $\text{🍷} = \text{🍷}$;
- (ii) $-\text{🍷} + \text{🍷} + \text{🍷} = \text{🍷}, -\text{🍷} + \text{🍷} + \text{🍷} = \text{🍷}, \text{🍷} + \text{🍷} + \text{🍷} = \text{🍷}, \text{🍷} - \text{🍷} = \text{🍷},$
- (iii) $\text{🍷} - \text{🍷} = \text{🍷}, \text{🍷} - \text{🍷} = \text{🍷}, \text{🍷} + \text{🍷} = \text{🍷}, \text{🍷} - \text{🍷} + \text{🍷} = \text{🍷}, \text{🍷} - \text{🍷} = \text{🍷}.$

Our goal is to show that the 🍷 in 🍷 will satisfies the relations (iv)-(v) of Proposition 3.1.1.

Using the reordered basis (3.1.2) and (3.2.4), we have

$$\begin{aligned} \text{🍷} &= \begin{pmatrix} \text{🍷} \\ \text{🍷} \\ \text{🍷} \end{pmatrix} \begin{pmatrix} \text{🍷} & \text{🍷} \end{pmatrix} - \text{🍷}, \quad \text{🍷} = \begin{pmatrix} \text{🍷} \\ \text{🍷} \\ \text{🍷} \end{pmatrix} \text{🍷}, \\ \text{🍷} &= \text{🍷} \begin{pmatrix} \text{🍷} & \text{🍷} \end{pmatrix}, \quad \text{🍷} = \text{🍷} - \text{🍷}. \end{aligned} \quad (3.3.1)$$

Denote

$$\text{🍷} := \text{🍷}, \quad \text{🍷} := \text{🍷}, \quad \text{🍷} := \text{🍷}, \quad \text{🍷} := \text{🍷}. \quad (3.3.2)$$

Apply (3.2.4) to (3.1.3), we get

$$\text{🍷} = (\text{🍷})(\text{🍷}), \quad \text{🍷} = (\text{🍷})(\text{🍷}).$$

Since $\text{🦉} = \text{🌱}$ and $\text{🌳} = \text{🌱}$, this implies

$$\text{🎵} = \text{🎵} \begin{pmatrix} -\text{🌳} \text{🦉} \text{🌱} \\ \text{🌳} \text{🦉} \text{🌱} \\ \text{🌳} \text{🦉} \text{🌱} \end{pmatrix}, \quad \text{🎵} = \text{🎵} \begin{pmatrix} \text{🌳} \text{🦉} \text{🌱} \\ \text{🌳} \text{🦉} \text{🌱} \\ -\text{🌳} \text{🦉} \text{🌱} \end{pmatrix}. \quad (3.3.3)$$

Moreover as $\text{🎵} = \text{🎵} \begin{pmatrix} \text{🌳} \text{🦉} \text{🌱} \\ \text{🌳} \text{🦉} \text{🌱} \\ \text{🌳} \text{🦉} \text{🌱} \end{pmatrix}$, by comparing the $\text{🦉}'$ row, we get $\text{🎵} = \text{🎵}$.

Lemma 3.3.1. In 🌱 , we have $\text{🎵} \text{🌱} + \text{🌱} \text{🎵} = -\text{🌱} \text{🌱}$, or equivalently

$$\text{🌱} = \text{🌱}, \text{🌱} = \text{🌱}, \text{🌱} = -\text{🌱} - \text{🌱}.$$

Proof. By (3.2.4), we have

$$\text{🎵} \text{🌱} + \text{🌱} \text{🎵} = \begin{pmatrix} \text{🌱} \text{🌱} & \text{🌱} \text{🌱} \\ \text{🌱} \text{🌱} & \text{🌱} \text{🌱} \end{pmatrix} = \begin{pmatrix} \text{🌱} \\ \text{🌱} \end{pmatrix} \begin{pmatrix} \text{🌱} & \text{🌱} \end{pmatrix}.$$

Therefore it suffices to verify that:

$$\begin{pmatrix} \text{🌳} \text{🦉} \text{🌱} \\ -\text{🌳} \text{🦉} \text{🌱} \end{pmatrix} \begin{pmatrix} \text{🌱} & \text{🌱} \end{pmatrix} + \begin{pmatrix} \text{🌱} \text{🌱} \\ \text{🌱} \text{🌱} \end{pmatrix} \begin{pmatrix} -\text{🌱} & \text{🌱} \end{pmatrix} = \text{🌱}.$$

Recall that $\text{🌳} = \text{🌱}$, by (3.2.6), we have:

$$\text{🌱} = \begin{cases} \text{🦉} \text{🌱} & \text{if } \text{🌱} \in [\text{🌱}, \text{🌱}] \cup [\text{🦉} - \text{🌱} + \text{🌱}, \text{🦉}]; \\ \text{🌱} \text{🌱} & \text{if } \text{🌱} \in [\text{🌱} + \text{🌱}, \text{🦉} - \text{🌱}]. \end{cases} \quad (3.3.4)$$

The desired relations now follow from (3.3.4) together with $\text{🎵} \text{🌱} = -\text{🌱} \text{🌱}$ and $\text{🌱} \text{🎵} = \text{🌱} \text{🌱}$ from (3.3.3). $\square$

Lemma 3.3.2. In 🌱 , the following relations are equivalent:

$$\begin{aligned} \text{🎵} \text{🌱} &= \text{🌱} \text{🌱}, \quad \text{🌱} \text{🌱} = \text{🌱} \text{🌱}, \quad \text{🌱} (\text{🌱}) = -(\text{🦉} - \text{🌱} - \text{🌱}) \text{🌱}, \\ \text{🌱} \text{🌱} &= \text{🌱} \text{🌱}, \quad \text{🌱} \text{🌱} = \text{🌱} \text{🌱}. \end{aligned}$$

Proof. We have

$$\text{🎵} \text{🌱} = \text{🌱} (\text{🌱} \text{🌱}) = \text{🌱} (\text{🌱} + \text{🌱} \text{🌱}) = \text{🌱} (\text{🌱}) + \text{🌱} (\text{🌱} \text{🌱}) + \text{🌱} (\text{🌱}) + \text{🌱} \text{🌱}.$$

Since $\text{🌱} (\text{🌱}) = \text{🌱} (\text{🌱})$ and $\text{🌱} + \text{🌱} = -\text{🌱} \text{🌱}$, we have

$$\text{🌱} (\text{🌱}) + \text{🌱} (\text{🌱}) + \text{🌱} \text{🌱} = (\text{🦉} - \text{🌱} - \text{🌱}) \text{🌱}.$$

Therefore,

$$\text{🎵} \text{🌱} = \text{🌱} (\text{🌱}) + (\text{🦉} - \text{🌱} - \text{🌱}) \text{🌱}.$$

We see that $\text{🎵} \text{🌱} = \text{🌱} \text{🌱}$ is equivalent to $\text{🌱} (\text{🌱}) = -(\text{🦉} - \text{🌱} - \text{🌱}) \text{🌱}$. The same argument applies to $\text{🌱} \text{🌱} = \text{🌱} \text{🌱}$. Now

$$\text{🎵} \text{🌱} \text{🌱} = \text{🌱} (\text{🌱} \text{🌱}) = \text{🌱} (\text{🌱} + \text{🌱} \text{🌱}) = \text{🌱} (\text{🌱}) + \text{🌱} \text{🌱} \text{🌱}.$$

We get the last two equivalences. $\square$

Finally, we have

Lemma 3.3.3. In 🌱 , the relations

$$-\text{🌱} \text{🌱} + \text{🌱} \text{🌱} = \text{🌱}, \quad \text{🌱} \text{🌱} + \text{🌱} \text{🌱} = \text{🌱}, \quad (3.3.5)$$

are equivalent to the relations

$$\text{🌱} = -\text{🌱} \text{🌱}, \quad \text{🌱} = \text{🌱} \text{🌱}.$$

$$\begin{aligned} \text{🌀} \text{🐰} &= \text{🍵} \text{🌳} \text{🌀} \text{🍀} \text{🏔️} \text{🌀} \text{🍊} = \text{🍵} \text{🌳} \text{🍀} \text{🌀} \text{🍊} \text{🍵} \text{🌳} \text{🍀} \text{🌀} = \begin{pmatrix} -\text{🎻} \text{🌱} \\ \text{🏔️} \text{🎻} \text{🍀} \end{pmatrix} \left(\text{🌳} \text{🐰} \text{🎻} \text{🌳} \text{🏔️} - \text{🍵} \text{🌳} \text{🐰} \text{🏔️} \right). \end{aligned}$$
$$\begin{pmatrix} \text{compass} \\ \text{rabbit} \end{pmatrix} = \begin{pmatrix} \text{candle} \\ \begin{pmatrix} - \\ \begin{pmatrix} \text{mountain} & \text{banjo} & \text{plant} & \text{tree} & \text{rabbit} & \text{mountain} \\ \text{mountain} & \text{banjo} & \text{clover} & \text{tree} & \text{rabbit} & \text{mountain} \end{pmatrix} \end{pmatrix} \end{pmatrix} = \begin{pmatrix} \text{candle} \\ \begin{pmatrix} - \\ \begin{pmatrix} \text{candle} & \text{wave} & \text{egg} \\ \text{candle} & \text{tree} & \text{egg} \end{pmatrix} \end{pmatrix} \end{pmatrix}.$$

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Proof. We want to show relation (iv) and (v) in Proposition 3.1.1 follows from the Kramer condition. We first verify this in the chart . The relations (iv) follows when we replace  by  -  and  by  - . The relations (v) follows from Lemma 3.3.1, 3.3.2 and 3.3.3.

$$\Lambda(\text{🕯️} - \text{🍰} \mid \text{🍷}) = \text{🥚}, \quad \Lambda(\text{🕯️} + \text{🍰} \mid \text{🍷}) = \text{🥚}.$$

Let $\text{🐎} \in \text{🏰}[\text{👉}] \setminus \text{☂️}$ be a point in the special fiber with image $\text{🐎}(\text{🐎}) \in \text{🏰}^\wedge[\text{👉}]$. Let $\text{☂️}^\wedge \subset \text{🏰}^\wedge[\text{👉}]$ be the open chart of $\text{🏰}^\wedge[\text{👉}]$ contains the worst point defined in the same way as ☂️ in §3.1. The subschemes $\text{☂️}^\circ \subseteq \text{☂️}^\wedge$ share the same underlying topological space. In particular, the reduced locus of the special fiber ☂️_2 can be written as the union of Schubert cells, all of them will specialize to the worst point.

Lemma 3.4.1. *In , the following relations hold:*

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Lemma 3.4.2. In $\text{🌂🍌}, \text{🍌🍌} = \text{🍌🍌}$ and $\text{🍌🍌}(\text{🍌🍌🍌🍌}) = \text{🍌🍌}$ implies $\text{🍌🍌🍌🍌} = \text{🍌🍌🍌🍌}$.

Proof. Since $\text{🍌🍌} = \text{🍌🍌}$, the condition $\text{🍌🍌🍌🍌} = \text{🍌🍌🍌🍌}$ is equivalent to $\text{🍌🍌} = \text{🍌🍌🍌🍌}$, by assumption. Now we have

$$\text{🍌🍌} = (\text{🍌🍌🍌🍌} - \text{🍌🍌🍌🍌}) = \text{🍌🍌🍌🍌🍌🍌🍌🍌} - \text{🍌🍌🍌🍌🍌🍌🍌🍌} + \text{🍌🍌🍌🍌} = \text{🍌🍌🍌🍌🍌🍌🍌🍌} - \text{🍌🍌🍌🍌🍌🍌🍌🍌}.$$

We conclude the following:

Theorem 3.4.3. When 🍌🍌 is not 🍌🍌 -modular, the naive splitting model is a closed subscheme of

$$\left[\text{🍌🍌} \times \text{🍌🍌} \times \text{🍌🍌}(\text{🍌🍌} \otimes \text{🍌🍌}) \right] \times \left[\text{🍌🍌} \times \text{🍌🍌} \times \text{🍌🍌}(\text{🍌🍌} \otimes \text{🍌🍌}) \right].$$

The open subscheme $\text{🍌🍌} \subset \text{🍌🍌}$ with coordinates 🍌🍌 of 🍌🍌 , and coordinates $\text{🍌🍌}, \text{🍌🍌}$ of 🍌🍌 , is cut out by the following relations:

$$(i) \text{🍌🍌} = -\text{🍌🍌} - \text{🍌🍌}, \text{🍌🍌} = -\text{🍌🍌} - \text{🍌🍌}, \text{🍌🍌} = \text{🍌🍌} - \text{🍌🍌};$$

$$(ii) \text{🍌🍌} = \text{🍌🍌}, \text{🍌🍌} = -\text{🍌🍌};$$

$$(iii) \text{🍌🍌} = \text{🍌🍌};$$

$$(iv) \text{🍌🍌} = \text{🍌🍌} = \text{🍌🍌};$$

$$(v) \text{🍌🍌} = \begin{pmatrix} \text{🍌🍌} & \text{🍌🍌} & \text{🍌🍌} \\ \text{🍌🍌} & \text{🍌🍌} & \text{🍌🍌} \\ \text{🍌🍌} & \text{🍌🍌} & \text{🍌🍌} \end{pmatrix} = \text{🍌🍌} - \text{🍌🍌}, \text{🍌🍌} = \begin{pmatrix} \text{🍌🍌} & \text{🍌🍌} & -\text{🍌🍌} \\ \text{🍌🍌} & \text{🍌🍌} & -\text{🍌🍌} \\ -\text{🍌🍌} & -\text{🍌🍌} & \text{🍌🍌} \end{pmatrix} = \text{🍌🍌} - \text{🍌🍌};$$

$$(vi) \begin{pmatrix} \text{🍌🍌} \\ \text{🍌🍌} \\ \text{🍌🍌} \end{pmatrix} \in \text{🍌🍌}, \begin{pmatrix} \text{🍌🍌} \\ \text{🍌🍌} \\ \text{🍌🍌} \end{pmatrix} \in \text{🍌🍌}.$$

Remark 3.4.4. By relation (v), the variables 🍌🍌 and 🍌🍌 are actually redundant.

Proof. The splitting model relations (3.2.4)-(3.2.6) are exactly equations (iv)-(vi). By Theorem 3.3.4, we are left to show that the relations in (LM1)-(LM4) are equivalent to (i)-(iii).

Since $\text{🍌🍌} + \text{🍌🍌} = -\text{🍌🍌}$, the relation $-\text{🍌🍌} + \text{🍌🍌} + \text{🍌🍌} = \text{🍌🍌}$ implies that

$$\text{🍌🍌} = \text{🍌🍌} - \text{🍌🍌} - \text{🍌🍌}.$$

This is exactly (i). The relations (ii) and (iii) hold by the previous discussion.

Conversely, suppose we have (i)-(iii). It is clear that (i) implies

$$\text{🍌🍌} = \text{🍌🍌}, \text{🍌🍌} = \text{🍌🍌}, \text{🍌🍌} = -\text{🍌🍌} - \text{🍌🍌}.$$

This then implies $-\text{🍌🍌} + \text{🍌🍌} + \text{🍌🍌} = \text{🍌🍌}$ by Lemma 3.3.1 and (i). Next, (ii) implies $-\text{🍌🍌} + \text{🍌🍌} = \text{🍌🍌}$ and $\text{🍌🍌} + \text{🍌🍌} = \text{🍌🍌}$ by Lemma 3.3.3. Moreover, (iii) and (iv) imply $\text{🍌🍌} - \text{🍌🍌} = \text{🍌🍌}$ by Lemma 3.4.2. The relations $\text{🍌🍌} - \text{🍌🍌} = \text{🍌🍌}, \text{🍌🍌} - \text{🍌🍌} = \text{🍌🍌}, \text{🍌🍌} + \text{🍌🍌} = \text{🍌🍌}$ follow from (v) by Lemma 3.4.1. Finally, (iii) implies that $\text{🍌🍌} - \text{🍌🍌} = \text{🍌🍌}$ since $\text{🍌🍌} = \text{🍌🍌}$ by Lemma 3.4.1. $\square$

3.4.5. In the remaining part of the section, we will prove Theorem 1.3.1. By Proposition 2.5.1, it suffices to prove the theorem in every affine chart $\text{🍌🍌} \subset \text{🍌🍌}$ of the naive splitting model corresponding to $\text{🍌🍌} = \text{🍌🍌} = \text{🍌🍌}$. By symmetry, it falls into four different classes:

- (i) $\text{🍌🍌} \in [\text{🍌🍌} + \text{🍌🍌}, \text{🍌🍌} - \text{🍌🍌}], \text{🍌🍌} \in [\text{🍌🍌}, \text{🍌🍌}] \cup [\text{🍌🍌} - \text{🍌🍌} + \text{🍌🍌}, \text{🍌🍌}].$
- (ii) $\text{🍌🍌}, \text{🍌🍌} \in [\text{🍌🍌}, \text{🍌🍌}] \cup [\text{🍌🍌} - \text{🍌🍌} + \text{🍌🍌}, \text{🍌🍌}].$
- (iii) $\text{🍌🍌}, \text{🍌🍌} \in [\text{🍌🍌} + \text{🍌🍌}, \text{🍌🍌} - \text{🍌🍌}].$
- (iv) $\text{🍌🍌} \in [\text{🍌🍌}, \text{🍌🍌}] \cup [\text{🍌🍌} - \text{🍌🍌} + \text{🍌🍌}, \text{🍌🍌}], \text{🍌🍌} \in [\text{🍌🍌} + \text{🍌🍌}, \text{🍌🍌} - \text{🍌🍌}].$

[illegible]

We will provide a detailed proof for case (i) as it is most interesting. The remaining cases will follow from similar computations and proofs.

(1) When 🗝️ + 🌱 ≤ 🦊 ≤ 🦋 - 🗝️ and 🌱 ≤ 🐶 ≤ 🗝️.

(2) When 🗝️ + 🌱 ≤ 🦊 ≤ 🦋 - 🗝️ and 🦋 - 🗝️ + 🌱 ≤ 🐶 ≤ 🦋.

We assume that 🗝️ is not 🍷-modular. In the affine chart 🌂 with 🦩 = 🌱, 🌳 = 🌿 for some 🗝️ + 🌱 ≤ 🐮 ≤ 🦋 - 🗝️ and 🌱 ≤ 🐶 ≤ 🗝️, we have:

-  =     ,   =    ,   =    , and   =  ;
-   = -      ,   =       ,   =     ;
-   =      ,   =      ,   = -      .

$$\left(\begin{array}{c} \text{green frog} \\ \text{basket} \\ \text{green frog} \end{array} \right) \in \left(\begin{array}{c} \text{brown cat} \\ \text{brown cat} \\ \text{brown cat} \end{array} \right) \Rightarrow \left(\begin{array}{c} \text{green frog} \\ \text{basket} \\ \text{green frog} \end{array} \right) = \text{green frog} \left(\begin{array}{c} \text{brown cat} \\ \text{brown cat} \\ \text{brown cat} \end{array} \right).$$
$$\left(\begin{array}{c} \text{pot, cat, plant} \\ \text{cat, dog} \\ \text{pot, dog, plant} \end{array} \right) \in \left[\text{img} \right] \Rightarrow \left(\begin{array}{c} \text{plant, plant} \\ \text{pot, cat, dog} \\ \text{pot, dog, plant} \end{array} \right) = \text{tree, cat} \left(\begin{array}{c} \text{plant, plant} \\ \text{plant, plant} \\ \text{plant, plant} \end{array} \right).$$

 =  ,  =  ,  =  .

3.5.2. Now we use the relation $\text{🌴} = \frac{1}{2} \text{🌿} \text{🍷}$. Since $\text{🌴} = \text{🌿} \text{🍷}$ and $\text{🌿} \text{🍷} = \text{🐯} \text{🌿}$, we have

$$\begin{pmatrix} \text{rabbit} & \text{musical note} & \text{cat} & \text{box} & \text{dragon} & \text{tiger} & \text{tree} & \text{mountain} \end{pmatrix} = \begin{pmatrix} \text{musical note} & \text{cat} & \text{box} \end{pmatrix} \begin{pmatrix} \text{dragon} & \text{rabbit} & \text{mountain} & \text{dragon} & \text{rabbit} & \text{tiger} & \text{tiger} & \text{tree} & \text{mountain} \end{pmatrix}.$$
$$\begin{array}{c} \text{rabbit} \quad \text{note} \quad \text{cat} \quad \text{box} \quad \text{dragon} \quad \text{tiger} \quad \text{tree} \\ \hline = \text{note} \quad \text{rabbit} \quad \text{box} \quad \left(\text{dragon} \quad \text{tree} \quad \text{mountain} \quad \text{dragon} \right) \quad \text{rabbit} \quad \text{tiger} \quad \text{tiger} \quad \text{tree} \end{array}$$
[illegible]

Proof. On the special fiber, $\text{🍌} = \text{🍌}$, and 🌂👤 is isomorphic to the factor ring $\text{🐞🍌}[\text{🐞🍌}, \text{🐞🍌}, \text{🐞🍌}, \text{🐞🍌}, \text{🐞🍌}]$ modulo the following relations

$$\text{🌳🍌} = \text{🍌}, \quad \text{🎵🍌} \left(\sum_{\text{🐞🍌}=\text{🍌}+\text{🍌}}^{\text{🐞🍌}-\text{🍌}} \text{🐞🍌} \text{🐞🍌} \text{🍌} \right) = \text{🍌}, \quad \text{🎵🍌} \left(\left(\text{🐞🍌} - \left(\sum_{\text{🐞🍌}=\text{🍌}+\text{🍌}}^{\text{🐞🍌}-\text{🍌}} \text{🐞🍌} \text{🐞🍌} \text{🍌} \right) \right) \right) = \text{🍌}.$$

On an irreducible component, the left hand side of each equation has at least one vanishing factor. By considering all possible combinations, we obtain the four irreducible components stated.

We now analyze these components individually: First, consider the locus defined by $\text{🎵🍌} = \text{🍌}$. This corresponds to the exceptional locus as $\text{🎵} = \text{🍌}$ and $\text{🎵} = \text{🍌}$ by Proposition 3.5.3, hence $\text{🌸} = \text{🍌}$ and $\text{🌸} = \text{🍌}$.

Let 🌳🍌 be the closed subscheme of 🌂👤 defined by $\text{🎵🍌} = \text{🍌}$ and $\text{🐞🍌} = \text{🍌}$. Then

$$\begin{aligned} \text{🌳🍌} &= \text{🐞🍌}[\text{🐞🍌}, \text{🐞🍌}, \text{🐞🍌}, \text{🐞🍌}, \text{🐞🍌}]/(\text{🌳🍌} - \text{🐞🍌}, \text{🐞🍌} - \text{🐞🍌}) \\ &\cong \begin{cases} \text{🐞🍌} \neq \text{🍌}; \\ \text{🐞🍌}(\text{🐞🍌} + \text{🐞🍌} + \dots, \text{🐞🍌}(\text{🐞🍌}-\text{🍌})/\text{🐞🍌}(\text{🐞🍌}+\text{🍌})/\text{🐞🍌} + \text{🐞🍌}) \subseteq \text{🐞🍌} & \text{🐞🍌} = \text{🍌}. \end{cases} \end{aligned}$$

Since $\text{🐞🍌} = \text{🍌}$, we have $\text{🐞🍌} = \text{🐞🍌} = \text{🍌}$ by Proposition 3.5.3. Hence $\text{🐞🍌}(\text{🍌}) = (\text{🍌})$ and $\text{🌳🍌} \subset \text{🐞🍌}$ by Proposition 2.5.4. Similar the component defined by $\text{🎵🍌} = \text{🍌}$ and $\text{🌳🍌} = \text{🍌}$ is a subscheme of 🐞🍌 .

Next, consider the subscheme of 🌂👤 defined by

$$\text{🐞🍌} = \text{🍌}, \quad \sum_{\text{🐞🍌}=\text{🍌}+\text{🍌}}^{\text{🐞🍌}-\text{🍌}} \text{🐞🍌} \text{🐞🍌} \text{🍌} = \text{🍌}.$$

It is isomorphic to

$$\text{🐞🍌}[\text{🐞🍌}, \text{🐞🍌}, \text{🐞🍌}, \text{🐞🍌}, \text{🐞🍌}]/(\text{🌳🍌} - \text{🐞🍌}, \text{🐞🍌} - \text{🐞🍌}, \sum_{\text{🐞🍌}=\text{🍌}+\text{🍌}}^{\text{🐞🍌}-\text{🍌}} \text{🐞🍌} \text{🐞🍌} \text{🍌}) \simeq \text{🐞🍌}.$$

By Proposition 2.5.4, $\text{🐞🍌} = \text{🍌}$ implies $\text{🐞🍌} = \text{🐞🍌} = (\text{🍌})$, hence $\text{🎵} = \text{🍌} = \text{🍌} = \text{🍌} = \text{🍌}$, in other words $\text{🐞🍌} = \text{🍌}$. Hence the image of this component under 🌂 lies in 🐞🍌 , see Remark 3.1.2. This component is not supported in the exceptional divisor, therefore it corresponds to $\widetilde{\text{🐞🍌}}$. Similarly, we obtain the corresponding equations for $\widetilde{\text{🐞🍌}}$. $\square$

Notice that 🌂👤 is not flat: the component in the integral ring defined by $\text{🎵🍌} = \text{🍌}$ cannot be lifted to the generic fiber since the special fiber is reduced and integrally we have $\text{🎵🍌} \text{🌳🍌} \left(\sum_{\text{🐞🍌}=\text{🍌}+\text{🍌}}^{\text{🐞🍌}-\text{🍌}} \text{🐞🍌} \text{🐞🍌} \text{🍌} \right) = \text{🍌}$. Consider the closed subscheme $\text{🌂}' \subset \text{🌂👤}$ defined by

$$\text{🐞🍌} - \text{🎵🍌} \left(\sum_{\text{🐞🍌}=\text{🍌}+\text{🍌}}^{\text{🐞🍌}-\text{🍌}} \text{🐞🍌} \text{🐞🍌} \text{🍌} \right). \quad (3.5.2)$$

It is isomorphic to

$$\begin{aligned}
 \text{🌂}' &= \frac{\text{🏠} [\text{🐼}, \text{🌿}, \text{🌿}, \text{🐼}, \text{🌿}, \text{🌿}, \text{🌿}, \text{🎵}, \text{🏠}]}{\text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}}, \\
 &\quad \text{🎵} \text{🌿} \text{🏠} \left(\sum_{\text{🌿}=\text{🌿}+\text{🌿}} \text{🌿} - \text{🌿} \right) - \text{🌿} \text{🌿}, \\
 &\quad \text{🌿} \text{🌿} - \text{🎵} \text{🌿} \left(\sum_{\text{🌿}=\text{🌿}+\text{🌿}} \text{🌿} - \text{🌿} \right) \\
 &\simeq \frac{\text{🏠} [\text{🐼}, \text{🌿}, \text{🌿}, \text{🐼}, \text{🌿}, \text{🌿}, \text{🌿}, \text{🎵}, \text{🏠}]}{\text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}}, \\
 &\quad \text{🌿} \text{🌿} - \text{🎵} \text{🌿} \left(\sum_{\text{🌿}=\text{🌿}+\text{🌿}} \text{🌿} - \text{🌿} \right)
 \end{aligned} \tag{3.5.3}$$

The special fiber $\text{🌂}'$ has three irreducible components, defined by

- $\sum_{\text{🌿}=\text{🌿}+\text{🌿}} \text{🌿} - \text{🌿} = \text{🍌}$,
- $\text{🌿} = \text{🍌}$,
- $\text{🎵} \text{🌿} = \text{🍌}$,

which corresponds to $\widetilde{\text{🌿}}, \widetilde{\text{🌿}}$, and 🌿 respectively. Therefore, it is the union

$$\text{🌂}' = \widetilde{\text{🌿}} \cup \widetilde{\text{🌿}} \cup \text{🌿}.$$

By substituting 🌿 (as 🌿 is invertible), we have the isomorphism

$$\text{🌂}' \simeq \frac{\text{🏠} [\text{🐼}, \text{🌿}, \text{🌿}, \text{🐼}, \text{🌿}, \text{🌿}, \text{🌿}, \text{🎵}, \text{🏠}]}{\text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}}. \tag{3.5.4}$$

It is Zariski locally isomorphic to an affine ring of the form

$$\frac{\text{🏠} [\text{🌿}, \text{🌿}, \text{🌿}, \text{🌿}, \text{🌿}, \text{🌿}, \text{🌿}, \text{🌿}, \text{🌿}]}{\text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}, \text{🌿} - \text{🌿}}.$$

Therefore it is flat over 🏠 with semi-stable reduction, and $\text{🌂}' = \text{🌂} \cap \text{🏠} \subset \text{🏠}$. To summarize, we have

Theorem 3.5.5. *With notation as above,*

(i) *the affine chart $\text{🌂}'$ is not flat over 🏠 . Moreover,*

$$\text{🌂}' = \text{🌂} \cap \text{🏠}$$

is flat over 🏠 with semi-stable reduction.

(ii) *Over the special fiber, we have*

$$\begin{aligned}
 \text{🌂} &= \widetilde{\text{🌿}} \cup \widetilde{\text{🌿}} \cup \text{🌿} \cup \text{🌿}; \\
 \text{🌂}' &= \widetilde{\text{🌿}} \cup \widetilde{\text{🌿}} \cup \text{🌿}.
 \end{aligned}$$

All four irreducible components are isomorphic to the affine space 🌿 . □

Remark 3.5.6. We have considered affine charts of class (i) when the index 🌿 is strongly non-special. It remains to consider the case when $\text{🌿} = \text{🌿} + \text{🌿}$ and $\text{🌿} = \text{🌿}$ and the case when $\text{🌿} = \text{🌿}$ and $\text{🌿} = \text{🌿} - \text{🌿}$.

Case 🌿 : $\text{🌿} = \text{🌿} + \text{🌿}$ and $\text{🌿} = \text{🌿}$. In this case, $\text{🌿} + \text{🌿} = \text{🌿} - \text{🌿} = \text{🌿} + \text{🌿} = \text{🌿}$. According to Proposition 3.6.2, the summation $\sum_{\text{🌿}=\text{🌿}+\text{🌿}} \text{🌿} - \text{🌿} = \text{🌿}$, which equals 🌿 since $\text{🌿} - \text{🌿} = \text{🍌}$.

[illegible]

Case $\text{🐰} : \text{🦋} = \text{🐰} + \text{🍄}$ and $\text{🐡} = \text{🍄} - \text{🌱}$. Here, $\text{🐡} + \text{🌱} = \text{🍄}$ and $\text{🦋} - \text{🐡} = \text{🍄} + \text{🌱}$, and $\text{🐰} = \text{🍄}$ or $\text{🍄} + \text{🌱}$. In Proposition 3.6.2, the summation $\sum_{\text{🐰}=\text{🍄}+\text{🌱}} \text{🦋} - \text{🐡} = \text{🍄} + \text{🌱}$. Since $\text{🦋} - \text{🐡} = \text{🌱}$, this term remains irreducible, and the discussions for the strongly non-special cases still apply. However, in the affine charts where $\text{🦋} - \text{🍄} = \text{🌱}$ and $\text{🦋} - \text{🍄} + \text{🌱} = \text{🌱}$, the irreducible component $\widetilde{\text{🦋} - \text{🐡}}$ correspond to $\text{🦋} - \text{🍄} + \text{🌱} = \text{🍄}$ and $\text{🦋} - \text{🍄} = \text{🍄}$ respectively. These are different irreducible components in the local model and splitting model. Only the affine charts with $\text{🐰} \in [\text{🌱}, \text{🐡}] \cup [\text{🦋} - \text{🐡} + \text{🌱}, \text{🦋}]$ includes both components. This relates to the fact that in this case, $\text{🦋} - \text{🐡}$ further decompose into two irreducible components, see [Luo25, Remark 4.3.6].

Proposition 3.6.1. *We have the following relations among , , , :*

- $\text{🍌} = \text{🌳} \text{🍌} \text{🌿}, \text{🍌} = \text{🍌} \text{🌿} \text{🍌}, \text{🍌} = \text{🌳} \text{🌿}.$ In particular $\text{🌳} \text{🍌} = \text{🌿};$
- $\text{🎸} = -\text{🔔} \text{🐼} \text{🏠} \text{🌿} \text{🌿}, \text{🎸} = \text{🍌} \text{🔔} \text{🏠} \text{🌿} \text{🌿} \text{🌿}, \text{🎸} = \text{🔔} \text{🐼} \text{🌳} \text{🌿} \text{🌿}.$ In particular, $\text{🔔} \text{🏠} = \text{🔔} \text{🐼} \text{🌳} \text{🌿} \text{🌿}.$
- $\text{🎸} = \text{🔔} \text{🐼} \text{🏠} \text{🌿} \text{🌿}, \text{🎸} = \text{🔔} \text{🐼} \text{🏠} \text{🌿} \text{🌿} \text{🌿} = -\text{🔔} \text{🐼} \text{🏠} \text{🌿} \text{🌿}.$ □

Apply $\text{🌴} = -\text{🌱}$ and $\text{🎸} = \text{🎻}$ to Proposition 3.6.1, we get

$$\begin{array}{c} \text{🐰} \text{🔔} \text{🐱} = \text{🔔} \text{🐱} \text{🐰} = \left(\sum_{\substack{\text{🐛} \text{🔑} \\ \text{🐛} \text{🔑} + \text{🌱}}} \text{🐛} \text{🔑} \text{🐱} \right) \quad \text{and} \quad \left(\sum_{\substack{\text{🐛} \text{🔑} \\ \text{🐛} \text{🔑} + \text{🌱}}} \text{🍷} \text{🔔} \text{🐱} \right) = \text{🐰} \text{🍷}. \end{array} \quad (3.6.1)$$

Proposition 3.6.2. *The affine chart  is isomorphic to the spectrum of the ring*

$$\cong \frac{\text{[House, Bird, ..., Butterfly, Bell, Box, Duck]}}{\left(\sum_{\text{Ant}=\text{Key}+\text{Plant}}^{\text{Butterfly}} - \text{Duck} - \text{Box} \right) - \left(\sum_{\text{Ant}=\text{Key}+\text{Plant}}^{\text{Butterfly}} - \text{Duck} - \text{Box} \right)}.$$

Over the special fiber, we have $\text{🌻} = \text{👦🏹🌳}$. By Proposition 3.6.1 we have $\text{👦🏹} = \text{🕒}$ which implies $\text{🧱👦🏹} = \text{💥👦🏹}$ by Proposition 2.5.4. We also get $\text{🕒🍊} = \text{👦🏹🌳} = \text{🥚}$. Hence, by Remark 3.1.2, we have $\text{🌳}(\text{🌂👦🏹}) \subset \text{⚡}$, see Remark 3.1.2.

Observe that $\text{bell} + \text{key} + \text{box}$ and $\text{bell} + \text{key} + \left(\sum_{\text{bug}} - \text{key} + \text{plant} + \text{snake} + \text{bug} + \text{snake} + \text{box} \right) - \text{rabbit}$ cannot both be zero on a geometric point of $\text{umbrella} + \text{dog} + \text{cat}$, as this would imply $-\text{rabbit} = \text{egg}$. Hence we have the decomposition

   = '   $\sqcup$    ,

where $\mathcal{U}_{\text{open}}$ is the open subscheme where $\mathcal{U}_{\text{open}}(\sum_{\text{open}} \mathcal{U}_{\text{open}} - \mathcal{U}_{\text{open}})$ is a unit and $\mathcal{U}'_{\text{open}}$ is the open subscheme where $\mathcal{U}_{\text{open}}$ is a unit. By (3.6.1), we have $\mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}}$ on $\mathcal{U}_{\text{open}}$. Hence

$$\mathcal{U}_{\text{open}} = \frac{\mathcal{U}_{\text{open}}[\mathcal{U}_{\text{open}}, \dots, \mathcal{U}_{\text{open}}, \mathcal{U}_{\text{open}}^{-1}]}{\mathcal{U}_{\text{open}} - \mathcal{U}_{\text{open}}, \mathcal{U}_{\text{open}}, -\mathcal{U}_{\text{open}}} \cong \mathcal{U}_{\text{open}}.$$

Since $\mathcal{U}_{\text{open}}$ is supported on the special fiber of $\mathcal{U}_{\text{open}}$, we see that $\mathcal{U}_{\text{open}}$ is not flat over $\mathcal{U}_{\text{open}}$.

Since $\mathcal{U}_{\text{open}}$ is a unit on $\mathcal{U}'_{\text{open}}$, we have $\mathcal{U}'_{\text{open}} = \mathcal{U}_{\text{open}}$ where

$$\mathcal{U}'_{\text{open}} = \frac{\mathcal{U}_{\text{open}}[\mathcal{U}_{\text{open}}, \dots, \mathcal{U}_{\text{open}}, \mathcal{U}_{\text{open}}^{-1}]}{\mathcal{U}_{\text{open}} - \mathcal{U}_{\text{open}}, \mathcal{U}_{\text{open}}(\sum_{\text{open}} \mathcal{U}_{\text{open}} + \mathcal{U}_{\text{open}}) - \mathcal{U}_{\text{open}}}.$$

Therefore, $\mathcal{U}'_{\text{open}}$ and $\mathcal{U}_{\text{open}}$ are the closed subschemes of $\mathcal{U}_{\text{open}}$ defined by the following two ideals respectively:

$$\mathcal{U}_{\text{open}} = \left(\mathcal{U}_{\text{open}} \left(\sum_{\text{open}} \mathcal{U}_{\text{open}} \right) - \mathcal{U}_{\text{open}} \right), \quad \mathcal{U}'_{\text{open}} = \left(\mathcal{U}_{\text{open}}, \mathcal{U}_{\text{open}} \right).$$

Over the special fiber, $\mathcal{U}_{\text{open}}(\mathcal{U}'_{\text{open}}) \subset \mathcal{U}_{\text{open}}(\mathcal{U}_{\text{open}})$ does not pass through the worst point: indeed, $\mathcal{U}_{\text{open}}$ is invertible which implies $\mathcal{U}_{\text{open}} \neq \mathcal{U}_{\text{open}}$ since $\mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}}$. Therefore, $\mathcal{U}'_{\text{open}} = \mathcal{U}_{\text{open}}$. On the other hand, the image $\mathcal{U}_{\text{open}}(\mathcal{U}_{\text{open}})$ is the worst point as $\mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}}$ on $\mathcal{U}_{\text{open}}(\mathcal{U}_{\text{open}})$. Therefore, $\mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}}$.

Theorem 3.6.3. *With notation as above and $\mathcal{U}_{\text{open}}$ is not in Yu's case.*

(i) *The affine chart $\mathcal{U}_{\text{open}}$ is a disjoint union*

$$\mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}} \sqcup \mathcal{U}'_{\text{open}},$$

where $\mathcal{U}_{\text{open}} \cong \mathcal{U}_{\text{open}}$ and

$$\mathcal{U}'_{\text{open}} \cong \frac{\mathcal{U}_{\text{open}}[\mathcal{U}_{\text{open}}, \dots, \mathcal{U}_{\text{open}}, \mathcal{U}_{\text{open}}^{-1}]}{\mathcal{U}_{\text{open}} - \mathcal{U}_{\text{open}}, \mathcal{U}_{\text{open}}(\sum_{\text{open}} \mathcal{U}_{\text{open}} + \mathcal{U}_{\text{open}}) - \mathcal{U}_{\text{open}}}. \quad (3.6.2)$$

In particular $\mathcal{U}_{\text{open}}$ is not flat over $\mathcal{U}_{\text{open}}$. Moreover

$$\mathcal{U}'_{\text{open}} = \mathcal{U}_{\text{open}} \cap \mathcal{U}_{\text{open}}(\mathcal{U}_{\text{open}})$$

is flat and smooth.

(ii) *Over the special fiber, we have identifications*

$$\mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}}, \quad \mathcal{U}'_{\text{open}} = \mathcal{U}_{\text{open}}.$$

Proof. It is clear that $\mathcal{U}'_{\text{open}}$ is flat, hence $\mathcal{U}'_{\text{open}} = \mathcal{U}_{\text{open}} \cap \mathcal{U}_{\text{open}}(\mathcal{U}_{\text{open}})$. The smoothness follows from the computation of the Jacobian matrix. $\square$

Remark 3.6.4. In Yu's case, i.e., when $\mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}}$ and $\mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}} - \mathcal{U}_{\text{open}}$, we have $\sum_{\text{open}} \mathcal{U}_{\text{open}} + \mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}} + \mathcal{U}_{\text{open}}$. Consequently, $\mathcal{U}_{\text{open}}$ further decompose into irreducible components defined by $\mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}}$ and $\mathcal{U}_{\text{open}} + \mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}}$, respectively. In this case, $\mathcal{U}'_{\text{open}}$ remains flat, but the special fiber now consists of two smooth irreducible components. This corresponds to the two irreducible components of $\mathcal{U}_{\text{open}}$ mentioned in Remark 3.5.6.

3.7. Affine charts of class (iii). In this case, we have $\mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}}$ for some $\mathcal{U}_{\text{open}} + \mathcal{U}_{\text{open}} \leq \mathcal{U}_{\text{open}}$, $\mathcal{U}_{\text{open}} \leq \mathcal{U}_{\text{open}} - \mathcal{U}_{\text{open}}$. At this point, we do not impose any restrictions on $\mathcal{U}_{\text{open}}$.

Proposition 3.7.1. *We have the following relations among $\mathcal{U}_{\text{open}}, \mathcal{U}_{\text{open}}, \mathcal{U}_{\text{open}}, \mathcal{U}_{\text{open}}$:*

- $\mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}}, \mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}}, \mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}}$, in particular $\mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}}$;
- $\mathcal{U}_{\text{open}} = -\mathcal{U}_{\text{open}}, \mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}}, \mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}}$;
- $\mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}}, \mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}}, \mathcal{U}_{\text{open}} = -\mathcal{U}_{\text{open}}$, in particular, $\mathcal{U}_{\text{open}} = \mathcal{U}_{\text{open}}$.

3.7.2. Applying Proposition 3.7.1 to $\text{🐛} = \text{🐛} + \text{🐛}$, we get

$$\text{🐛} = \sum_{\text{🐛} = \text{🐛} + \text{🐛}}^{\text{🐛} - \text{🐛}} \text{🐛} = \text{🐛} \left(\sum_{\text{🐛} = \text{🐛} + \text{🐛}}^{\text{🐛} - \text{🐛}} \text{🐛} \right).$$

By the same computation as in the proof of Proposition 3.5.3, one deduces:

Proposition 3.7.3. *The affine chart 🐛 is isomorphic to the spectrum of the ring*

$$\text{🐛} \simeq \frac{\text{🏠}[\text{🌳}, \dots, \text{🌳}, \text{🎵}, \text{🎵}, \text{🌳}^{-1}]}{\text{🌳} - \text{🌳}, \text{🎵} \left(\sum_{\text{🐛} = \text{🐛} + \text{🐛}}^{\text{🐛} - \text{🐛}} \text{🌳} \right) - \text{🐛}}.$$

□

Assume 🐛 is not almost 🐛 -modular, i.e., not where $\text{🐛} = \text{🐛} + \text{🐛}$ and $\text{🐛} = \text{🐛}$. Denote $\text{🐛} = \text{🐛} + \text{🐛}$. The special fiber 🐛 consists of two irreducible components defined by

$$\text{🌳} : \text{🎵} = \text{🐛}, \quad \text{🌳}' : \sum_{\text{🐛} = \text{🐛} + \text{🐛}}^{\text{🐛} - \text{🐛}} \text{🌳} = \text{🐛}.$$

It is straightforward to see that 🌳 and $\text{🌳}'$ are smooth over 🐛 .

By Proposition 3.7.1 we have $\text{🌳} = \text{🐛}$, $\text{🌳}' = \text{🐛}$ on the special fiber, hence $\text{🌳} \subset \text{🌳}'$ by Proposition 2.5.4. Recall that $\text{🌳} = \text{🌳}$, hence we have $\text{🌳} = \text{🌳} = \text{🌳} = \text{🌳} = \text{🌳}$. Therefore, we have $\text{🌳} \subset \text{🌳}'$, see Remark 3.1.2. Moreover, since $\text{🎵} = \text{🐛}$ implies $\text{🌳} = \text{🐛}$, the image $\text{🌳}(\text{🌳})$ is the worst point. Hence

$$\text{🌳}' = \text{🌳}, \quad \text{🌳} = \text{🌳}.$$

Theorem 3.7.4. *When 🐛 is not almost 🐛 -modular.*

(i) *The affine chart $\text{🐛}' := \text{🐛}$ is flat and regular, with semi-stable reduction.*

(ii) *The special fiber equals*

$$\text{🐛}' = \text{🌳}' \cup \text{🌳}.$$

□

Remark 3.7.5. When 🐛 is almost 🐛 -modular, we have $\sum \text{🌳} = \text{🌳}$, hence we only have 🌳 in this chart.

3.8. **Affine charts of class (iv).** We consider the case when $\text{🌳} \leq \text{🐛} \leq \text{🐛}$ and $\text{🐛} + \text{🌳} \leq \text{🐛} \leq \text{🐛} - \text{🐛}$, the other cases in this class follow from the same computation. In this case, as $\text{🌳} = \text{🌳}$, we have

$$\begin{pmatrix} \text{🌳} \\ \text{🐛} \\ \text{🌳} \end{pmatrix} \in \begin{pmatrix} \text{🐛} \\ \text{🐛} \\ \text{🐛} \end{pmatrix} \text{ implies } \begin{pmatrix} \text{🌳} \\ \text{🐛} \\ \text{🌳} \end{pmatrix} = \text{🐛} \begin{pmatrix} \text{🌳} \\ \text{🐛} \\ \text{🌳} \end{pmatrix}.$$

Similarly, since $\text{🌳} = \text{🌳}$, we have

$$\begin{pmatrix} \text{🐛} \\ \text{🐛} \\ \text{🐛} \end{pmatrix} \in \begin{pmatrix} \text{🌳} \\ \text{🌳} \\ \text{🌳} \end{pmatrix} \text{ implies } \begin{pmatrix} \text{🐛} \\ \text{🐛} \\ \text{🐛} \end{pmatrix} = \text{🐛} \begin{pmatrix} \text{🌳} \\ \text{🌳} \\ \text{🌳} \end{pmatrix}.$$

In particular, we get $\text{🐛} = \text{🌳} = \text{🌳}$, hence 🐛 is invertible in class (iii). Therefore, 🐛 is contained in the generic fiber, hence is smooth, hence regular. In this case we define $\text{🐛}' := \text{🐛}$.

Combining all the discussions, we finish the proof of Theorem 1.3.1. □

4. RESOLUTION OF SINGULARITY

In this section we assume 🗝️ is not 🍷-modular. The main result of this section is Theorem 4.0.1 below. After proving the theorem, we will identify the exceptional divisor 🌸🌱 with a blow-up of 🏠🌱 in Proposition 4.3.2.

Theorem 4.0.1. *For 🗝️ not 🍷-modular, the morphism*

$$\text{🌱} : \text{🏠}_{[\text{🗝️}]} \longrightarrow \text{🏠}_{[\text{🗝️}]}$$

is the blow-up of 🏠_{[\text{🗝️}]} along the worst point.

4.1. Defining equations of the blow-up. We define 🏠_{[\text{🗝️}]} to be the blow up of 🏠_{[\text{🗝️}]} over the worst point 🌀. By Theorem 1.3.1 and the universal property of the blow up, we have a unique natural morphism 🏠_{[\text{🗝️}]} \rightarrow 🏠_{[\text{🗝️}]} defined over 🏠_{[\text{🗝️}]} . By Proposition 2.5.1, 🌱 is an isomorphism outside the worst point 🌀. Therefore, it suffices to consider an affine neighborhood of 🌀 as in Proposition 3.1.1. We denote 🌱 to be the blow-up of 🌱 over the worst point.

Denote by 🌱 = 🌱 the affine ring of 🌱, which is the quotient algebra of 🏠_{[\text{🗝️}]} modulo the relations in Proposition 3.1.1. The worst point 🌀 is defined by the ideal 🌀 = (🌀, 🍷) \subset 🌱. Then the blow-up of 🌱 along 🌀 is

$$\text{🌱} = \text{🌱} \oplus \text{🌱} \text{ with } \text{🌱} = \bigoplus_{\text{🌱} \geq \text{🌀}} \text{🌱} \text{ with } \text{🌱} = \text{🌱}.$$

We define 🏠 be another 🦋 \times 🦋 matrix of independent variables:

$$\text{🏠} = \begin{pmatrix} \text{📚} & \text{🍷} & \text{📚} \\ \text{📚} & \text{🏠} & \text{📚} \\ \text{📚} & \text{📚} & \text{📚} \end{pmatrix}, \quad \text{🌱} = \begin{pmatrix} \text{🗝️} & \text{🌱} & -\text{📚} \\ \text{🗝️} & \text{🌱} & -\text{📚} \\ -\text{📚} & -\text{🌱} & \text{📚} \end{pmatrix}. \quad (4.1.1)$$

We think of the variables in 🏠 as variables in 🌱 \subset 🌱. Parallel to §3, we denote

$$\text{🏠}_{\text{🌱}} = \begin{pmatrix} \text{🗝️} & \text{📚} \\ \text{📚} & \text{📚} \end{pmatrix}, \text{🏠}_{\text{🌱}} = \begin{pmatrix} \text{🗝️} \\ \text{📚} \end{pmatrix}, \text{🏠}_{\text{🌱}} = \begin{pmatrix} \text{🗝️} & \text{📚} \end{pmatrix}.$$

Define 🌱 = 🌱[\text{🏠}, \text{🍷}]/\text{🌱} where 🌱 is the ideal of 🌱[\text{🏠}, \text{🍷}] (homogeneous with respect to 🏠 and 🍷) generated by

$$\begin{aligned} & \text{🌱} - \text{🌱}, \text{🌱} - \text{🌱}, \text{🌱} - \text{🌱}, \text{🌱} - \text{🌱}, \\ & \text{🌱} + \text{🌱} - \text{🌱}, \text{🌱} + \text{🌱} - \text{🌱}, \text{🌱} + \text{🌱} - \text{🌱}, \text{🌱} + \text{🌱} - \text{🌱}, \\ & \wedge (\text{🌱} + \text{🌱}), \wedge (\text{🌱} - \text{🌱}), \\ & \text{📚} - \text{📚}, \text{📚} - \text{📚}, \text{📚} + \text{🌱} - \text{📚}, \text{📚} - \text{📚}, \text{📚} + \text{🌱} - \text{📚}, \\ & \text{🗝️} + \text{🌱} - \text{📚}, \text{📚} + \text{🌱} - \text{📚}, \text{📚} - \text{🌱} - \text{📚}, \text{📚} - \text{🌱} - \text{📚}, \\ & \text{🌱}(\text{🌱}) = -(\text{🌱} - \text{🌱} - \text{🌱}). \end{aligned} \quad (4.1.2)$$

Let 🌱 := 🌱. By definition 🌱 is a closed subscheme of 🌱. Suppose 🌱 \leq 🌱, 🌱 \leq 🌱. Define 🌱 to be the open subscheme of 🌱 where 🌱 + 🌱 \neq 🌱 where 🌱 is the Kronecker delta.

Lemma 4.1.1. *🌱 = 🌱[\text{🏠}, \text{🍷}]/\text{🌱} can be covered by \{\text{🌱} \mid \text{🌱} \leq 🌱, \text{🌱} \leq 🌱\}.*

Proof. By the relation 🌱(\text{🌱}) = -(\text{🌱} - \text{🌱} - \text{🌱}), the lemma follows from the same argument as that of [Shi22, Lemma A.1]. $\square$

On 🌱 we can assume 🌱 + 🌱 = 🌱. Define 🌱 = (\text{🌱}, \dots, \text{🌱}) to be the 🌱-th column of 🌱 + 🌱, i.e.,

$$\text{🌱} = \text{🌱} + \text{🌱}.$$

Then $\text{🦊} = \text{🌱}$. Also define

$$\text{👑} = \text{👑} + \text{🍪}, \text{ and } \text{🔔} = \text{🔔} + \text{🍪}.$$

In particular $\text{🦊} = \text{🌱}$.

Lemma 4.1.2. *We have*

$$\text{👑} + \text{🍪} = \text{🦊}, \quad (4.1.3)$$

and

$$\text{🌸} + \text{🍪} = \text{🦊}. \quad (4.1.4)$$

Proof. By the first line of relations in 4.1.2, we have

$$(\text{🌸} + \text{🍪})(\text{👑} + \text{🍪}) = (\text{🌸} + \text{🍪})(\text{👑} + \text{🍪}). \quad (4.1.5)$$

Relation (4.1.3) follows directly from the wedge condition for 👑 as

$$(\text{👑} + \text{🍪})(\text{👑} + \text{🍪}) - (\text{👑} + \text{🍪})(\text{👑} + \text{🍪}) = \text{🍪},$$

and $\text{👑} + \text{🍪} = \text{🌱}$ on 🍪 . Now by (4.1.5), we have

$$\text{🌸} + \text{🍪} = (\text{🌸} + \text{🍪})(\text{👑} + \text{🍪}).$$

Combining with (4.1.3), we have

$$\text{🌸} + \text{🍪} = (\text{🌸} + \text{🍪})\text{🦊}.$$

So in order to show (4.1.4), it suffices to show that

$$(\text{🌸} + \text{🍪})\text{🦊} = \text{🌱},$$

or equivalently

$$(\text{🌸} + \text{🍪})(\text{👑} + \text{🍪}) = \text{🌸} + \text{🍪}.$$

But this again follows from (4.1.5). $\square$

Similarly, by the relation

$$\text{👑} + \text{🍪} = \text{🦊}(\text{👑} + \text{🍪}),$$

we know that if $\text{👑} + \text{🍪} = \text{🌱}$, then

$$\text{👑} + \text{🍪} = \pm \text{🌱}.$$

Define

$$\text{🌱} = \begin{cases} \text{👑} + \text{🍪} & \text{if } \text{👑} + \text{🍪} = \text{🌱}, \\ -(\text{👑} + \text{🍪}) & \text{if } \text{👑} + \text{🍪} = -\text{🌱}. \end{cases}$$

Also define

$$\text{🔔} = \text{👑} + \text{🍪}, \text{ and } \text{🎵} = \text{🔔} + \text{🍪}.$$

Then $\text{🌱} = \text{🌱}$ and $\text{🔔} = \pm \text{🌱}$. By the same proof as that of Lemma 4.1.2, we have

$$\text{👑} + \text{🍪} = \text{🦊}, \quad (4.1.6)$$

and

$$\text{🌸} + \text{🍪} = \text{🦊}. \quad (4.1.7)$$

4.2. Proof of the main theorem. In this section, we will prove Theorem 4.0.1.

Recall from Theorem 3.4.3 that $\mathcal{U}_{\text{sc}}^{\text{sc}}$ is the open subscheme $\mathcal{U}_{\text{sc}}^{\text{sc}}(\mathcal{U}_{\text{sc}}^{\text{sc}})$ of $\mathcal{U}_{\text{sc}}^{\text{sc}}$, and we have a natural morphism

$$\mathcal{U}_{\text{sc}}^{\text{sc}} : \mathcal{U}_{\text{sc}}^{\text{sc}} \longrightarrow \mathcal{U}_{\text{sc}}^{\text{sc}} \hookrightarrow \widetilde{\mathcal{U}}, \quad (4.2.1)$$

where $\mathcal{U}_{\text{sc}}^{\text{sc}} \rightarrow \mathcal{U}_{\text{sc}}^{\text{sc}}$ is the unique morphism obtained from the universal property of the blow-up $\mathcal{U}_{\text{sc}}^{\text{sc}}$. Recall §3.4.5 that $\mathcal{U}_{\text{sc}}^{\text{sc}}$ is covered by $\{\mathcal{U}_{\text{sc}}^{\text{sc}}\}$ and Lemma 4.1.1, that $\widetilde{\mathcal{U}}$ is covered by $\{\mathcal{U}_{\text{sc}}^{\text{sc}}\}$. The main result of this section is the following.

Theorem 4.2.1. (i) For charts of cases (i)-(iii), the morphism $\mathcal{U}_{\text{sc}}^{\text{sc}}$ induces isomorphisms

$$\mathcal{U}_{\text{sc}}^{\text{sc}} : \mathcal{U}_{\text{sc}}^{\text{sc}} \rightarrow \mathcal{U}_{\text{sc}}^{\text{sc}}$$

for all indices $\mathcal{U}_{\text{sc}}^{\text{sc}}$ and $\mathcal{U}_{\text{sc}}^{\text{sc}}$.

(ii) The charts of case (iv) lie over the generic fiber. In particular, $\mathcal{U}_{\text{sc}}^{\text{sc}} : \mathcal{U}_{\text{sc}}^{\text{sc}} \rightarrow \mathcal{U}_{\text{sc}}^{\text{sc}}$ is an isomorphism.

The proof of Theorem 4.2.1 will base on an explicit computation on local charts, which is the main content of this subsection. Let us assume this for now.

Proof of Theorem 4.0.1. By the definition of $\widetilde{\mathcal{U}}$ in (4.1.2) and [Luo25, Theorem 8.0.1], $\mathcal{U}_{\text{sc}}^{\text{sc}}$ is an isomorphism over the generic fiber $\mathcal{U}_{\text{sc}}^{\text{sc}}$. By Lemma 4.1.1 and Theorem 4.2.1, the composition $\mathcal{U}_{\text{sc}}^{\text{sc}}$ is étale. We have the following diagram of proper maps:

$$\begin{array}{ccc} \mathcal{U}_{\text{sc}}^{\text{sc}} & \xrightarrow{\quad} & \mathcal{U}_{\text{sc}}^{\text{sc}} \\ \downarrow & \searrow & \\ \mathcal{U}_{\text{sc}}^{\text{sc}} & & \end{array}$$

Therefore, the base change $\mathcal{U}_{\text{sc}}^{\text{sc}} \rightarrow \mathcal{U}_{\text{sc}}^{\text{sc}}$ is also proper, so is the composition $\mathcal{U}_{\text{sc}}^{\text{sc}}$. By [Sta, 03WS], $\mathcal{U}_{\text{sc}}^{\text{sc}}$ is étale and therefore locally quasi-finite. By [Sta, 02LS], it is finite. Since it is an isomorphism on the generic fiber, the finite étale morphism $\mathcal{U}_{\text{sc}}^{\text{sc}}$ is an isomorphism. It follows that the morphism $\mathcal{U}_{\text{sc}}^{\text{sc}} \rightarrow \mathcal{U}_{\text{sc}}^{\text{sc}}$ is an isomorphism. By Proposition 2.5.1, the morphism $\mathcal{U}_{\text{sc}}^{\text{sc}} \rightarrow \mathcal{U}_{\text{sc}}^{\text{sc}}$ is an isomorphism. $\square$

Now we come to the proof of Theorem 4.2.1. We will separate them into the following four cases:

- (i) $\mathcal{U}_{\text{sc}}^{\text{sc}} \in [\mathcal{U}_{\text{sc}}^{\text{sc}}, \mathcal{U}_{\text{sc}}^{\text{sc}}] \cup [\mathcal{U}_{\text{sc}}^{\text{sc}} - \mathcal{U}_{\text{sc}}^{\text{sc}} + \mathcal{U}_{\text{sc}}^{\text{sc}}, \mathcal{U}_{\text{sc}}^{\text{sc}}]$, $\mathcal{U}_{\text{sc}}^{\text{sc}} \in [\mathcal{U}_{\text{sc}}^{\text{sc}} + \mathcal{U}_{\text{sc}}^{\text{sc}}, \mathcal{U}_{\text{sc}}^{\text{sc}} - \mathcal{U}_{\text{sc}}^{\text{sc}}]$.
- (ii) $\mathcal{U}_{\text{sc}}^{\text{sc}}, \mathcal{U}_{\text{sc}}^{\text{sc}} \in [\mathcal{U}_{\text{sc}}^{\text{sc}}, \mathcal{U}_{\text{sc}}^{\text{sc}}] \cup [\mathcal{U}_{\text{sc}}^{\text{sc}} - \mathcal{U}_{\text{sc}}^{\text{sc}} + \mathcal{U}_{\text{sc}}^{\text{sc}}, \mathcal{U}_{\text{sc}}^{\text{sc}}]$.
- (iii) $\mathcal{U}_{\text{sc}}^{\text{sc}}, \mathcal{U}_{\text{sc}}^{\text{sc}} \in [\mathcal{U}_{\text{sc}}^{\text{sc}} + \mathcal{U}_{\text{sc}}^{\text{sc}}, \mathcal{U}_{\text{sc}}^{\text{sc}} - \mathcal{U}_{\text{sc}}^{\text{sc}}]$.
- (iv) $\mathcal{U}_{\text{sc}}^{\text{sc}} \in [\mathcal{U}_{\text{sc}}^{\text{sc}}, \mathcal{U}_{\text{sc}}^{\text{sc}}] \cup [\mathcal{U}_{\text{sc}}^{\text{sc}} - \mathcal{U}_{\text{sc}}^{\text{sc}} + \mathcal{U}_{\text{sc}}^{\text{sc}}, \mathcal{U}_{\text{sc}}^{\text{sc}}]$, $\mathcal{U}_{\text{sc}}^{\text{sc}} \in [\mathcal{U}_{\text{sc}}^{\text{sc}} + \mathcal{U}_{\text{sc}}^{\text{sc}}, \mathcal{U}_{\text{sc}}^{\text{sc}} - \mathcal{U}_{\text{sc}}^{\text{sc}}]$.

We will construct the isomorphism $\mathcal{U}_{\text{sc}}^{\text{sc}}$ defined over $\mathcal{U}_{\text{sc}}^{\text{sc}}$ directly in cases (i)-(iii).

As in (3.3.2), denote

$$\mathcal{U}_{\text{sc}}^{\text{sc}} := \mathcal{U}_{\text{sc}}^{\text{sc}}, \quad \mathcal{U}_{\text{sc}}^{\text{sc}} := \mathcal{U}_{\text{sc}}^{\text{sc}}, \quad \mathcal{U}_{\text{sc}}^{\text{sc}} := \mathcal{U}_{\text{sc}}^{\text{sc}}, \quad \mathcal{U}_{\text{sc}}^{\text{sc}} := \mathcal{U}_{\text{sc}}^{\text{sc}}. \quad (4.2.2)$$

Apply relations (4.1.4) and (4.1.7) to (3.1.3) (which still holds as it is a relation on $\mathcal{U}_{\text{sc}}^{\text{sc}}$), we get

$$\mathcal{U}_{\text{sc}}^{\text{sc}} = (\mathcal{U}_{\text{sc}}^{\text{sc}})(\mathcal{U}_{\text{sc}}^{\text{sc}}), \quad \mathcal{U}_{\text{sc}}^{\text{sc}} = (\mathcal{U}_{\text{sc}}^{\text{sc}})(\mathcal{U}_{\text{sc}}^{\text{sc}}). \quad (4.2.3)$$

Since $\mathcal{U}_{\text{sc}}^{\text{sc}} = \mathcal{U}_{\text{sc}}^{\text{sc}}$ and $\mathcal{U}_{\text{sc}}^{\text{sc}} = \mathcal{U}_{\text{sc}}^{\text{sc}}$, this implies

$$\mathcal{U}_{\text{sc}}^{\text{sc}} = \mathcal{U}_{\text{sc}}^{\text{sc}}(\mathcal{U}_{\text{sc}}^{\text{sc}}) = \mathcal{U}_{\text{sc}}^{\text{sc}} \begin{pmatrix} -\mathcal{U}_{\text{sc}}^{\text{sc}} \\ \mathcal{U}_{\text{sc}}^{\text{sc}} \\ \mathcal{U}_{\text{sc}}^{\text{sc}} \end{pmatrix}, \quad \mathcal{U}_{\text{sc}}^{\text{sc}} = \mathcal{U}_{\text{sc}}^{\text{sc}}(\mathcal{U}_{\text{sc}}^{\text{sc}}) = \mathcal{U}_{\text{sc}}^{\text{sc}} \begin{pmatrix} \mathcal{U}_{\text{sc}}^{\text{sc}} \\ \mathcal{U}_{\text{sc}}^{\text{sc}} \\ -\mathcal{U}_{\text{sc}}^{\text{sc}} \end{pmatrix}. \quad (4.2.4)$$

Moreover as $\mathcal{U}_{\text{sc}}^{\text{sc}} = \mathcal{U}_{\text{sc}}^{\text{sc}}$, by comparing the $\mathcal{U}_{\text{sc}}^{\text{sc}}$'s row, we get $\mathcal{U}_{\text{sc}}^{\text{sc}} = \mathcal{U}_{\text{sc}}^{\text{sc}}$.

Similarly denote

$$\mathcal{U}_{\text{sc}}^{\text{sc}} := \mathcal{U}_{\text{sc}}^{\text{sc}}, \quad \mathcal{U}_{\text{sc}}^{\text{sc}} := \mathcal{U}_{\text{sc}}^{\text{sc}}.$$

Apply relations (4.1.3) and (4.1.6) to the relation $\text{👤} = \text{🐉} \text{👤} \text{🐉}$, we get

$$\text{🐉} \text{👤} = (\text{🐉} \text{👤}) (\text{🐉} \text{👤}), \quad \text{👤} \text{🐉} = (\text{🐉} \text{👤}) (\text{🐉} \text{👤}).$$

Since $\text{🐉} = \text{🌱}$ and $\text{👤} = \text{🌱}$, this implies

$$\text{👤} = \text{👤} (\text{🐉} \text{👤}) = \text{👤} \begin{pmatrix} -\text{🐉} \text{👤} \\ \text{🐉} \text{👤} \\ \text{🐉} \text{👤} \end{pmatrix}, \quad \text{👤} = \text{👤} (\text{🐉} \text{👤}) = \text{👤} \begin{pmatrix} \text{🐉} \text{👤} \\ \text{🐉} \text{👤} \\ -\text{🐉} \text{👤} \end{pmatrix}. \quad (4.2.5)$$

Moreover as $\text{🐉} \text{👤} = \text{👤} \text{🐉}$, by comparing the 👤 's row, we get $\text{👤} \text{👤} = \pm \text{🌱}$.

From (4.1.3), we get

$$\text{👤} + \text{👤} = \text{👤} \text{👤}.$$

Comparing this with the last equation of (4.1.2), we get

$$\text{👤} = \text{🌱} \text{👤} \text{👤}. \quad (4.2.6)$$

4.2.2. *Case (i).* We focus on $\text{👤} - \text{👤} + \text{🌱} \leq \text{👤} \leq \text{👤}$ and $\text{👤} + \text{🌱} \leq \text{👤} \leq \text{👤} - \text{👤}$. The case $\text{🌱} \leq \text{👤} \leq \text{👤}$ and $\text{👤} + \text{🌱} \leq \text{👤} \leq \text{👤} - \text{👤}$ can be dealt with in the same way.

(1) By (4.2.4), we have

$$\text{👤} \text{🌱} = -\text{👤} \text{👤} \text{👤}, \quad \text{👤} \text{🌱} = \text{👤} \text{👤} \text{👤}, \quad \text{👤} \text{🌱} = \text{👤} \text{👤} \text{👤}.$$

and by (4.2.5), we have

$$\text{👤} = -\text{👤} \text{👤} \text{👤}, \quad \text{👤} = \text{👤} \text{👤} \text{👤}, \quad \text{👤} = \text{👤} \text{👤} \text{👤}. \quad (4.2.7)$$

In particular

$$\text{👤} = \text{👤} \text{👤} \text{👤} = \text{👤} \text{👤}, \quad \text{👤} = \text{👤} \text{👤} \text{👤} = \text{👤} \text{👤}.$$

(2) Next, we want to express 👤 in terms of 👤 and 👤 . From Theorem 3.4.3(v), we have $\text{👤} = \text{👤} \text{👤} \text{👤} - \text{👤} \text{👤}$, $\text{👤} = \text{👤} \text{👤} \text{👤} - \text{👤} \text{👤}$. By one of the strengthened spin relations $\text{👤} = -\text{👤} \text{👤} \text{👤} - \text{👤} \text{👤}$, we have $\text{👤} \text{👤} \text{👤} = -\text{👤} \text{👤} \text{👤}$, by comparing the 👤 's rows, we obtain

$$\text{👤} \text{👤} = -\text{👤} \text{👤} \text{👤}$$

From (4.2.4), we have

$$\text{👤} \text{👤} = \text{👤} \text{👤} \text{👤}.$$

Moreover, from Theorem 3.4.3(v), we have $\text{👤} = \text{👤} \text{👤} \text{👤}$, $\text{👤} = -\text{👤} \text{👤} \text{👤}$. Apply one of the strengthened spin conditions $\text{👤} = \text{👤}$, we get $\text{👤} \text{👤} \text{👤} = -\text{👤} \text{👤} \text{👤}$. By comparing the 👤 's rows, we deduce that

$$\text{👤} \text{👤} = -\text{👤} \text{👤} \text{👤}.$$

We apply the same arguments to 👤 , we can get:

$$\text{👤} \text{🌱} = -\text{👤} \text{👤} \text{👤}, \quad \text{👤} \text{🌱} = \text{👤} \text{👤} \text{👤} = \text{👤} \text{👤} \text{👤}, \quad \text{👤} \text{🌱} = -\text{👤} \text{👤} \text{👤}. \quad (4.2.8)$$

(3) From equations (4.1.3), (4.1.6), and expression (4.1.1), we have $\text{👤} = \text{👤} \text{👤} \text{👤} = -\text{👤} \text{👤} \text{👤}$. Applying (4.2.8), we obtain $\text{👤} \text{👤} \text{👤} = \text{👤} \text{👤} \text{👤}$. Comparing the 👤 -th row of this equation yields

$$\text{👤} \text{👤} = \text{👤} \text{👤} \text{👤}.$$

Similarly, using the same equations and expression, we have $\text{👤} + \text{👤} = \text{👤} \text{👤} \text{👤} = -\text{👤} \text{👤} \text{👤}$. Applying (4.2.8), we get $\text{👤} \text{👤} \text{👤} = \text{👤} \text{👤} \text{👤}$. Comparing the 👤 -th row gives:

$$\text{👤} \text{👤} = \text{👤} \text{👤} \text{👤}.$$

(3) Now we express 🔪 and 🐘 in terms of 🌿 and 🎵 . By (4.2.4), we have

$$\text{🔪} = \text{🔔} \text{🐘} \text{🌿}, \quad \text{🔪} = \text{🔔} \text{🐘} \text{🌿}, \quad \text{🔪} = -\text{🔔} \text{🐘} \text{🌿}.$$

By (4.2.5), we get $\text{🔔} = \text{🐘} \text{🌿} = \text{🐘} \text{🌿}$ and

$$\text{🐘} = \text{🐘} \text{🌿} = \text{🌿}, \quad \text{🐘} = \text{🐘} \text{🌿} = \text{🌿}, \quad \text{🐘} = -\text{🐘} \text{🌿} = -\text{🌿}.$$

Or equivalently $\text{🐘} = \text{🌿}$.

(4) We express 🐘 in terms of 🌿 . By (4.2.5), we get $\text{🐘} = \text{🔔} \text{🐘} \text{🌿} = \text{🌿}$ or equivalently $\text{🐘} = \text{🌿}$. Combining with (4.2.10), we get

$$\text{🐘} = \text{🌿}, \quad \text{🐘} = \text{🌿}, \quad \text{🐘} = \text{🌿}.$$

In particular $\text{🌿} = \text{🌿} = \text{🌿}$ or equivalently $\text{🌿} = \text{🌿}$.

(5) From (4.2.6), we get

$$\text{🍌} = \frac{\text{🌿}}{\text{🐘}} \text{🌿} = \frac{\text{🌿}}{\text{🐘}} \text{🌿}.$$

(6) To summarize we have express all the variables in terms of 🌿 and 🔔 , and 🐘 is a unit. Now consider the relation $\text{🔔} = \frac{\text{🌿}}{\text{🐘}}$, using the facts that $\text{🔔} = \text{🌿} \text{🐘}$ and $\text{🌿} = \text{🌿} \text{🐘}$, we have

$$\text{🔔} = \text{🌿} \text{🐘} = \text{🌿} \text{🐘}.$$

By comparing the $(\text{🐘} + \text{🌿} - \text{🐘})$ -th entry, we have

$$\text{🔔} = \text{🔔} \text{🐘} \text{🌿} \left(\sum_{\text{🐘} + \text{🌿}} \text{🐘} \text{🌿} \right).$$

Canceling $\text{🔔} \text{🐘}$ (recall $\text{🔔} \text{🐘} \text{🌿} = \text{🌿}$), we have

$$\text{🔔} = \text{🔔} \text{🐘} \text{🌿} \left(\sum_{\text{🐘} + \text{🌿}} \text{🐘} \text{🌿} \right).$$

Using the same argument as in §3.4, we can see that all the relations in (4.1.2) are satisfied. Comparing with the affine chart 🌿 in Theorem 3.6.3, we see that $\text{🔔} \text{🐘} \text{🌿} \cong \text{🌿}$ for $\text{🐘} = \text{🌿}$ and $\text{🌿} = \text{🌿}$.

4.2.4. *Case (iii).* We focus on 🔔 when $\text{🐘} + \text{🌿} \leq \text{🐘}$, $\text{🌿} \leq \text{🐘} - \text{🐘}$.

(1) We express 🔔 in terms of 🐘 and 🔪 . By (4.2.4), we have

$$\text{🔔} = -\text{🎵} \text{🐘} \text{🌿}, \quad \text{🔔} = \text{🎵} \text{🐘} \text{🌿}, \quad \text{🔔} = \text{🎵} \text{🐘} \text{🌿}.$$

(2) We express 🐘 in terms of 🌿 , by (4.2.5) we get

$$\text{🐘} = -\text{🔔} \text{🐘} \text{🌿}, \quad \text{🐘} = \text{🔔} \text{🐘} \text{🌿}, \quad \text{🐘} = \text{🔔} \text{🐘} \text{🌿}.$$

In this case $\text{🔔} \text{🐘} \text{🌿} = \text{🐘} \text{🌿} = \text{🌿}$, to summarize we have

$$\text{🐘} = \text{🌿}.$$

(3) We express 🔪 in terms of 🐘 and 🎵 . Recall that $\text{🌿} = \text{🌿}$. By the relation $\text{🌿} + \text{🐘} = \text{🔪}$, and comparing terms in the following strengthened spin relations:

$$\text{🌿} = -\text{🐘} \text{🌿}, \quad \text{🌿} = \text{🐘} \text{🌿}, \quad \text{🌿} = \text{🐘} \text{🌿},$$

we get

$$\text{🔪} = \text{🐘} \text{🎵} \text{🌿}, \quad \text{🔪} = \text{🎵} \text{🐘} \text{🌿}, \quad \text{🔪} = -\text{🐘} \text{🎵} \text{🌿}.$$

(4) Next we want to express 🐘 in terms of 🌿 . Using the relations $\text{🐘} + \text{🌿} = \text{🔪}$, $\text{🌿} = -\text{🐘} \text{🌿}$, $\text{🌿} = \text{🌿}$ and $\text{🌿} = \text{🌿}$, arguing exactly the same way as above, we have

$$\text{🐘} = \text{🐘} \text{🌿}, \quad \text{🐘} = \text{🐘} \text{🌿}, \quad \text{🐘} = -\text{🐘} \text{🌿}.$$

Since $\text{🏠🐱} = \text{🌱}$, we have $\text{🐱🌳} = \text{🏠🐱} = \text{🌱}$. In particular 🐱🌳 and 🌳🐱 are units. So we have

$$\text{🐘🌱} = \text{🍲🌳🐱🌱}, \quad \text{🐘🐱} = \text{🌳🐱🌱}, \quad \text{🐘🌳} = -\text{🍲🌳🐱🌱}. \quad (4.2.11)$$

(5) We express 🌱 in terms of 🐱 . By (4.2.5), we get $\text{🐘} = \text{🐱🌳🐱} = \text{🐱🌳}$ or equivalently $\text{🌱} = \text{🐱🌳🐘}$. Combining with (4.2.11), we get

$$\text{🌱🌱} = \text{🍲🌳🐱🌱}, \quad \text{🌱🐱} = \text{🌳🐱🌱}, \quad \text{🌱🌳} = -\text{🍲🌳🐱🌱}.$$

In particular $\text{🐱🌳} = \text{🌳🐱} = \text{🌱}$ or equivalently $\text{🐱🌳🐱} = \text{🌱}$. So

$$\text{🌱🌱} = \text{🍲🐱🌳🐱}, \quad \text{🌱🐱} = \text{🐱🌳🐱}, \quad \text{🌱🌳} = -\text{🍲🐱🌳🐱}.$$

(6) From (4.2.6), we get

$$\text{🍪} = \frac{\text{🌱}}{\text{🐱}} \text{🐱🌳🐱} = \frac{\text{🌱}}{\text{🐱}} \text{🌳🐱}.$$

(7) To summarize we have express all the variables in terms of 🐱 and 🌳 , and 🌳 is a unit. Consider the relation $\text{🔍}(\text{🍪}) = -(\text{🐱} - \text{🐱🌳} - \text{🐱})$, this is equivalent to $\text{🔍}(\text{🐱🌳} - \text{🐱} - \text{🐱}) = -(\text{🐱} - \text{🐱🌳} - \text{🐱})$, hence $\text{🔍}(\text{🐱🌳}) = \text{🐱}$. Using the relation $\text{🎵} = \text{🎵🌳🐱}$, we have

$$\text{🎵} \left(\sum_{\text{🐱}=\text{🌱}+\text{🌱}} \text{🌳} \right) = \text{🐱}.$$

Using the same argument as in §3.4, we can see that all the relations in (4.1.2) are satisfied. Comparing with Proposition 3.7.3, we see that $\text{🐱🌳} \cong \text{🐱}'$ for $\text{🐱} = \text{🐱}$ and $\text{🌳} = \text{🐱}$.

4.2.5. *Case (iv).* We focus on 🐱 when $\text{🌱} \leq \text{🐱} \leq \text{🔑}$ and $\text{🔑} + \text{🌱} \leq \text{🐱} \leq \text{🐱} - \text{🔑}$. The case $\text{🐱} - \text{🔑} + \text{🌱} \leq \text{🐱} \leq \text{🐱}$ and $\text{🔑} + \text{🌱} \leq \text{🐱} \leq \text{🐱} - \text{🔑}$ can be dealt with in the same way. By the relation $\text{🍪} = \text{🍲🌳🐱}$, we know that

$$\text{🐘} = \text{🍲🌳🐱}.$$

Since $\text{🐱} = \text{🌱}$, we have

$$\text{🍲🌳🐱} = \text{🐱} = \text{🌱}.$$

This shows that 🍲 is a unit and 🌳🐱 is supported on the generic fiber.

4.2.6. We can now conclude the proof of Theorem 4.2.1. In cases (i)-(iii), we have constructed an isomorphism $\text{🐱} : \text{🐱}' \rightarrow \text{🐱}$. By (3.2.4) and (4.1.4), these morphisms preserve the matrix 🌱 , hence are defined over 🐱 . Moreover, over 🐱 , we can define

$$\text{🐱} := \text{🐱} \begin{pmatrix} -\text{🌳🐱} \\ \text{🌳🐱} \\ \text{🌳🐱} \end{pmatrix},$$

as in (4.2.5) and obtain global sections $\text{🍪} := \frac{\text{🌱}}{\text{🐱}} \text{🐱🌳🐱}$ and $\text{🍲} (\text{🌱} \leq \text{🐱}, \text{🐱} \leq \text{🐱})$ of the line bundle $\text{🐱}(\text{🌱})$ over 🐱 where

$$\text{🍲} := -\text{🍪} + \text{🌳🐱}.$$

Via the morphisms 🐱 , these sections agree with the corresponding sections of the invertible sheaf $(\text{🐱} \cdot \text{🌱})^{-\text{🌱}}$ over 🐱 . This shows that the morphisms 🐱 are indeed induced by the morphism 🐱 in (4.2.1) (see for example the proof of [Har77, Proposition 7.14]). This finishes the proof. $\square$

4.3. Exceptional divisor. To the end of the section, we describe the exceptional divisor of the blow up. Consider the 🏠🌿 -vector space $\text{🏠🌿} = \text{🏠🌿} \otimes \text{🌿}$. Recall that we have the following standard 🏠🌿 -basis of 🏠🌿 ,

$$\text{🏠🌿}^{-1} \otimes \text{🌿}, \dots, \text{🏠🌿}^{-1} \otimes \text{🌿}, \text{🏠🌿}^{-1} \otimes \text{🌿}, \dots, \text{🏠🌿} \otimes \text{🌿}.$$

Recall we have maps $\text{🏠🌿} : \text{🏠🌿} \rightarrow \text{🏠🌿}$ and $\text{🏠🌿} : \text{🏠🌿} \rightarrow \text{🏠🌿}$. Define

$$\text{🏠🌿} = (\text{🏠🌿}) = \text{🏠🌿} \{ \text{🏠🌿}^{-1} \otimes \text{🌿}, \dots, \text{🏠🌿}^{-1} \otimes \text{🌿} \} \cong (\text{🏠🌿}) \text{🌿},$$

and

$$\begin{aligned} \text{🏠🌿} &= (\text{🏠🌿}) = (\text{🏠🌿}) \\ &= \text{🏠🌿} \{ \text{🏠🌿}^{-1} \otimes \text{🌿}, \dots, \text{🏠🌿}^{-1} \otimes \text{🌿}, \text{🏠🌿}^{-1} \otimes \text{🌿}, \dots, \text{🏠🌿} \otimes \text{🌿} \}. \end{aligned}$$

Then we have an exact sequence

$$\text{🏠} \rightarrow \text{🏠🌿} \rightarrow \text{🏠🌿} \rightarrow \text{🏠} \quad (4.3.1)$$

Denote by $\text{🏠🌿} = \text{🏠🌿}[\text{🌿}, \dots, \text{🌿}]$ the projective space of lines of the 🏠🌿 , 🏠🌿 the projective space of 🏠🌿 , and 🏠🌿 the projective space of 🏠🌿 . Here we use the homogeneous coordinate ring $\text{🏠🌿}[\text{🌿}, \dots, \text{🌿}]$ with respect to the standard basis. From the exact sequence (4.3.1), we get a monomorphism $\text{🏠🌿} \hookrightarrow \text{🏠🌿}$.

Lemma 4.3.1. *The subvariety of $\text{🏠🌿} \times \text{🏠🌿}$ characterized by*

$$\text{🏠} := \{ (\text{🏠}, \text{🏠}) \in \text{🏠🌿} \times \text{🏠🌿} \mid \text{🏠}(\text{🏠}) \subset \text{🏠} \}$$

is the blow-up of 🏠🌿 along 🏠🌿 .

Proof. Write homogeneous coordinates on $\text{🏠}(\text{🏠})$ as $[\text{🏠} : \dots : \text{🏠}, \text{🏠} : \dots : \text{🏠}]$, where the 🏠 's are coordinates on the quotient 🏠🌿 . Then $\text{🏠}(\text{🏠}) = \text{🏠}(\text{🏠}, \dots, \text{🏠}) \subset \text{🏠}(\text{🏠)$. Now let $[\text{🏠} : \dots : \text{🏠}]$ be coordinates on $\text{🏠}(\text{🏠)$. The condition $\text{🏠}(\text{🏠}) \subset \text{🏠}$ is exactly (A local chart computation will show that this defined a reduced closed subscheme)

$$\text{🏠} - \text{🏠} = \text{🏠} \quad \text{for all } \text{🏠}.$$

But the blow-up of $\text{🏠}(\text{🏠})$ along $\text{🏠}(\text{🏠}) = \text{🏠}(\text{🏠}, \dots, \text{🏠})$ is the closure of the graph of the rational map

$$\text{🏠}(\text{🏠}) \dashrightarrow \text{🏠}(\text{🏠}), \quad [\text{🏠} : \dots : \text{🏠}, \text{🏠} : \dots : \text{🏠}] \mapsto [\text{🏠} : \dots : \text{🏠}].$$

Inside $\text{🏠}(\text{🏠}) \times \text{🏠}(\text{🏠)$, this graph closure is also cut out by the equations $\text{🏠} - \text{🏠} = \text{🏠}$ for all 🏠 . Hence 🏠 and the blow-up are the same closed subscheme. $\square$

We denote the blow-up of 🏠🌿 along 🏠🌿 by $\text{🏠🌿}(\text{🏠🌿})$.

Proposition 4.3.2. *The exceptional divisor 🏠🌿 of the blow-up $\text{🏠🌿} \rightarrow \text{🏠🌿}$ is isomorphic to $\text{🏠🌿}(\text{🏠🌿})$.*

Proof. We have seen that there exists an open cover

$$\left\{ \text{🏠} \mid \text{🏠} \in [\text{🏠} + \text{🏠}, \text{🏠} - \text{🏠}], \text{🏠} \in [\text{🏠}, \text{🏠}] \cup [\text{🏠} - \text{🏠} + \text{🏠}, \text{🏠}] \right\}$$

of 🏠 . We will show that on each 🏠 there is a natural embedding $\text{🏠} \rightarrow \text{🏠🌿}(\text{🏠🌿})$.

First we assume that $\text{🏠} + \text{🏠} \leq \text{🏠} \leq \text{🏠} - \text{🏠}$. This corresponds to charts of type (ii) as in §3.7. In this case we know that

$$\text{🏠} = \text{🏠}[\text{🏠}, \dots, \text{🏠}] / (\text{🏠} - \text{🏠}) \cong \text{🏠},$$

Secondly we assume that $\text{🌱} \leq \text{🐶} \leq \text{🔑}$. As before the case $\text{🦋} - \text{🔑} + \text{🌱} \leq \text{🐶} \leq \text{🦋}$ is exactly the same, so is omitted. This corresponds to charts of type (iv) as in §3.5. In this case we know that

Obviously the images of the above embeddings cover $\text{img}(\text{embed}_{\text{img}}) \cup \text{img}(\text{embed}_{\text{img}}^{-1})$ and these embeddings agree on overlaps. This finishes the proof of the proposition. $\square$

When $\text{🌍} = \text{🔬🌍}$ is perfect, a nilpotent 🏠🌊 -display is equivalent to a relative Dieudonné module $\text{🏠}((\text{🌀}))$ over 🌈 (🌍) with the action of a 👤 -linear operator 🌊 and a $\text{👤}^{-1}\text{-}\text{🌱}$ -linear operator 🌿 , such that $\text{🌊🌿} = \text{🌿🌊} = \text{🍷🏠}$.

5.2. RZ spaces. For the remaining part of the paper, assume $\mathbb{F}/\mathbb{F}_0$ is a ramified quadratic extension of $\mathbb{F}_0$ -adic fields. Fix uniformizers $\varpi \in \mathbb{F}$ and $\varpi_0 \in \mathbb{F}_0$ such that $\varpi^2 = \varpi_0$.

Definition 5.2.1. (i) Let $\mathbb{X}$ be a $\mathbb{G}_m$ -scheme. A *hermitian $\mathbb{G}_m$ -module of type μ and dimension d* is a triple $(\mathcal{H}, \mathcal{F}, \mathcal{G})$ consisting of a strict biformal $\mathbb{G}_m$ -module $\mathcal{H}$ of height μ and dimension d over $\mathbb{X}$, a homomorphism

$$\mathcal{F}: \mathcal{H} \rightarrow \mathcal{H}(\varpi),$$

and a *relative polarization*

$$\mathcal{G}: \mathcal{H} \rightarrow \mathcal{H}(\varpi_0),$$

subject to the following constraints:

- the Rosati involution on $\mathcal{H}(\varpi)$ attached to $\mathcal{G}$ induces the nontrivial Galois automorphism on $\mathbb{F}$; and
- $\mathcal{H} \subset \mathcal{H}(\varpi_0)$ has height μ .

(ii) A hermitian $\mathbb{G}_m$ -module is *admissible* if it satisfies the following condition: denote by $\mathcal{H}(\varpi)$ and $\mathcal{H}(\varpi_0)$ the respective de Rham homology of $\mathcal{H}$ and $\mathcal{H}(\varpi_0)$. Since $\mathcal{H} \subset \mathcal{H}(\varpi_0)$ is contained in $\mathcal{H}(\varpi_0)$ and of rank d , there is a unique (necessarily $\mathbb{G}_m$ -linear) isogeny $\mathcal{G}$ such that the composite

$$\mathcal{H} \xrightarrow{\mathcal{G}} \mathcal{H}(\varpi_0) \xrightarrow{\mathcal{F}} \mathcal{H}(\varpi)$$

is $\mathcal{H}(\varpi)$, and the induced diagram

$$\mathcal{H}(\varpi) \xrightarrow{\mathcal{F}} \mathcal{H}(\varpi_0) \xrightarrow{\mathcal{G}} \mathcal{H}(\varpi)$$

then extends periodically to a polarized chain of $\mathbb{G}_m$ -modules $\mathcal{H}(\varpi)$, comp. (2.1.4). By [RZ96, Th. 3.16], étale-locally on $\mathbb{X}$ there exists an isomorphism of polarized chains

$$[\dots \xrightarrow{\mathcal{G}} \mathcal{H}(\varpi_0) \xrightarrow{\mathcal{F}} \mathcal{H}(\varpi) \xrightarrow{\mathcal{G}} \mathcal{H}(\varpi_0) \xrightarrow{\mathcal{F}} \dots] \xrightarrow{\sim} [\dots \xrightarrow{\mathcal{F}} \mathcal{H}(\varpi) \xrightarrow{\mathcal{G}} \mathcal{H}(\varpi_0) \xrightarrow{\mathcal{F}} \dots],$$

which in particular gives an isomorphism of $\mathbb{G}_m$ -modules

$$\mathcal{H}(\varpi) \xrightarrow{\sim} \mathcal{H}(\varpi_0) \otimes \mathcal{H}(\varpi_0). \quad (5.2.1)$$

The admissible condition we impose is that upon identifying $\mathcal{H}(\varpi)$ with a submodule of $\mathcal{H}(\varpi_0) \otimes \mathcal{H}(\varpi_0)$ via (5.2.1), it defines a point in the local model $\mathcal{M}_{\mathbb{F}_0}$.

(iii) A *splitting structure* on a hermitian $\mathbb{G}_m$ -module $(\mathcal{H}, \mathcal{F}, \mathcal{G})$ of dimension d and type μ is a pair of two locally $\mathbb{G}_m$ -direct summands of rank one,

$$\mathcal{H}(\varpi) \subseteq \mathcal{H}(\varpi_0) \subseteq \mathcal{H}(\varpi), \quad \mathcal{H}(\varpi_0) \subseteq \mathcal{H}(\varpi_0) \subseteq \mathcal{H}(\varpi_0),$$

subject to the following constraints:

- the morphisms between Hodge filtrations induced by the polarization carry the one additional filtration into the other:

$$\mathcal{F}(\mathcal{H}(\varpi)) \subseteq \mathcal{F}(\mathcal{H}(\varpi_0)), \quad \mathcal{G}(\mathcal{H}(\varpi_0)) \subseteq \mathcal{G}(\mathcal{H}(\varpi)).$$

- (*Kr mer condition*) if $d > 0$ is even and $\mu = d$, then the condition states that $\mathcal{H}(\varpi) = (\mathcal{F}(\mathcal{H}(\varpi)) - \mathcal{H}(\varpi_0)) \cap \mathcal{H}(\varpi)$. For the other cases, it requires

$$\begin{aligned} (\mathcal{F}(\mathcal{H}(\varpi)) - \mathcal{H}(\varpi_0)) \cap \mathcal{H}(\varpi) &\subseteq \mathcal{H}(\varpi), & (\mathcal{F}(\mathcal{H}(\varpi)) + \mathcal{H}(\varpi_0)) \cap \mathcal{H}(\varpi) &= (\varpi); \\ (\mathcal{F}(\mathcal{H}(\varpi)) - \mathcal{H}(\varpi_0)) \cap \mathcal{H}(\varpi_0) &\subseteq \mathcal{H}(\varpi_0), & (\mathcal{F}(\mathcal{H}(\varpi)) + \mathcal{H}(\varpi_0)) \cap \mathcal{H}(\varpi_0) &= (\varpi_0). \end{aligned}$$

Remark 5.2.2. (i) The admissible condition can be replaced by the strengthened spin condition, see [Luo25].

(ii) Let $(\otimes, \otimes, \otimes, \otimes)$ be any hermitian $\mathcal{H}$ -module. The action $\otimes : \mathcal{H} \rightarrow \otimes(\otimes)$ induces an action $\otimes : \mathcal{H} \rightarrow \otimes(\otimes)$. However throughout the paper unless otherwise stated, in order to make $\otimes$ have the same signature as $\otimes$ and have a natural definition of special cycles $\otimes(\otimes)$ (see below), we use the conjugate action $\overline{\otimes}$ of $\otimes$.

See also [RZ96, Definition 3.21(v)].

Definition 5.2.3. (i) A hermitian $\mathcal{H}$ -module $(\otimes, \otimes, \otimes, \otimes)$ over $\mathcal{H}$ is called the *framing object* if the relative Dieudonné module is isoclinic. See [LR25, Proposition 5.1.3] for a full classification of the framing objects.

(ii) Let $(\otimes, \otimes, \otimes, \otimes)$ be a framing object of type $\otimes$ and dimension $\otimes$. We define the *canonical RZ space* $\otimes$ over $\mathcal{H}$ as the formal scheme parametrizing isomorphism classes of quadruples $(\otimes, \otimes, \otimes, \otimes)$, where:

- $(\otimes, \otimes, \otimes, \otimes)$ is an admissible hermitian $\mathcal{H}$ -module of type $\otimes$;
- $\otimes$, called the *framing* with framing object $(\otimes, \otimes, \otimes, \otimes)$, is an $\mathcal{H}$ -linear quasi-isogeny of height $\otimes$

$$\otimes : \otimes \times \otimes \rightarrow \otimes \times \otimes,$$

such that $\otimes(\otimes \times \otimes) = \otimes \times \otimes$.

(iii) Fix a framing object $(\otimes, \otimes, \otimes, \otimes)$. We define the *naive splitting RZ space* $\otimes$ over $\mathcal{H}$ parametrizing the collection of data

$$(\otimes, \otimes, \otimes, \otimes, \otimes, \otimes, \otimes),$$

where $(\otimes, \otimes, \otimes, \otimes, \otimes, \otimes, \otimes)$ is a hermitian $\mathcal{H}$ -module of type $\otimes$ with splitting structure, and where $\otimes$ is a *framing* with the fixed framing object.

(iv) We define the *splitting RZ space* $\otimes$ over $\mathcal{H}$ as the flat closure of $\otimes$, i.e. closed formal subscheme defined by the ideal sheaf $\otimes[\otimes] \subseteq \otimes$ for $\otimes = \otimes$ ($\otimes$ -power torsion elements of the structure sheaf).

By Theorem 1.3.1 and Theorem 1.4.1, combine with the local model diagram, we immediately have

Proposition 5.2.4. (i) *There is natural chain of maps:*

$$\otimes \hookrightarrow \otimes \longrightarrow \otimes, \quad (5.2.2)$$

where the morphism on the left is a closed immersion.

(ii) *The splitting RZ space $\otimes$ is a flat formal scheme with semi-stable reduction.*

Proof. Part (i) is straightforward, part (ii) follows from the local model diagram for $\otimes$, which is obtained from the local model diagram for $\otimes$ via the modification [Pap00, Prop. 2.3] and the uniqueness of the flat closure. $\square$

5.3. The line bundle of modular forms. In this subsection, we define two natural line bundles over the naive splitting RZ space.

Let $\otimes$ be the Lubin-Tate $\mathcal{H}$ -module over $\mathcal{H}$, which is the unique strict $\mathcal{H}$ -module over $\mathcal{H}$. There is a unique lifting (the canonical lifting) $\otimes$ of Lubin-Tate $\mathcal{H}$ -module $\otimes$ over $\mathcal{H}$, equipped with its framing $\otimes : \otimes \times \otimes \xrightarrow{\sim} \otimes$ and its principal polarization $\otimes$ lifting $\otimes(\otimes)$.

Let $\otimes \leq \otimes \leq \otimes$ be any integer. Let $\otimes$ be a $\mathcal{H}$ -scheme over $\otimes$ and $\otimes \in \otimes(\otimes)$ be a $\otimes$ -point. In the remaining part of the paper, we can assume $\otimes = \otimes$ defined by the closed embedding $\otimes \hookrightarrow \otimes$. Recall that the de Rham homology $\otimes(\otimes)$ and $\otimes(\otimes)$ defined in §5.1 are vector bundles over $\otimes$ of rank $\otimes$. These fit into short exact sequences

$$\otimes \rightarrow \otimes(\otimes) \rightarrow \otimes(\otimes) \rightarrow \otimes(\otimes) \rightarrow \otimes, \quad \otimes \rightarrow \otimes(\otimes) \rightarrow \otimes(\otimes) \rightarrow \otimes(\otimes) \rightarrow \otimes.$$

Similarly, the de Rham homology of $\mathcal{H} = \mathcal{H} \times_{\mathcal{H}} \mathcal{H}$ determines a short exact sequence

$$\mathcal{O} \longrightarrow \mathcal{H}(\mathcal{H}) \longrightarrow \mathcal{H}(\mathcal{H}) \longrightarrow \mathcal{O} \longrightarrow \mathcal{O}$$

of vector bundles on $\mathcal{H}$.

From the moduli description, the actions $\mathcal{H} : \mathcal{H} \rightarrow \mathcal{H}(\mathcal{H})$ and $\mathcal{H} : \mathcal{H} \rightarrow \mathcal{H}(\mathcal{H})$ and $\mathcal{H} : \mathcal{H} \rightarrow \mathcal{H}(\mathcal{H})$ induce actions of $\mathcal{H}$ on all of these vector bundles over $\mathcal{H}$, and the above short exact sequences are $\mathcal{H}$ -linear.

By definition, we have a perfect alternating pairing

$$\langle \cdot, \cdot \rangle : \mathcal{H}(\mathcal{H}) \times \mathcal{H}(\mathcal{H}) \longrightarrow \mathcal{O},$$

which is compatible with the action $\mathcal{H} : \mathcal{H} \rightarrow \mathcal{H}(\mathcal{H})$ and $\mathcal{H} : \mathcal{H} \rightarrow \mathcal{H}(\mathcal{H})$, in the sense that

$$\langle \mathcal{H} \cdot \mathcal{H}, \mathcal{H} \cdot \mathcal{H} \rangle = \langle \mathcal{H}, \mathcal{H} \rangle$$

for all $\mathcal{H} \in \mathcal{H}$ and all local sections $\mathcal{H}$ (resp. $\mathcal{H}$) of $\mathcal{H}(\mathcal{H})$ (resp. $\mathcal{H}(\mathcal{H})$).

The local direct summands $\mathcal{H}(\mathcal{H}) \subset \mathcal{H}(\mathcal{H})$ and $\mathcal{H}(\mathcal{H}) \subset \mathcal{H}(\mathcal{H})$ are perpendicular, hence there are two induced perfect pairings

$$\langle \cdot, \cdot \rangle : \mathcal{H}(\mathcal{H}) \times \mathcal{H}(\mathcal{H}) \longrightarrow \mathcal{O}, \text{ and } \langle \cdot, \cdot \rangle : \mathcal{H}(\mathcal{H}) \times \mathcal{H}(\mathcal{H}) \longrightarrow \mathcal{O}. \quad (5.3.1)$$

By virtue of the moduli problem defining $\mathcal{H}$, we have distinguished local direct summands $\mathcal{H}(\mathcal{H}) \subset \mathcal{H}(\mathcal{H})$ and $\mathcal{H}(\mathcal{H}) \subset \mathcal{H}(\mathcal{H})$ of rank $\mathcal{H}$ on $\mathcal{H}$. By taking dual via (5.3.1), we have local direct summands $\mathcal{H}(\mathcal{H}) \subset \mathcal{H}(\mathcal{H})$ and $\mathcal{H}(\mathcal{H}) \subset \mathcal{H}(\mathcal{H})$ of rank $\mathcal{H} - \mathcal{H}$ respectively. Let

$$\mathcal{H} := \mathcal{H} \left(\mathcal{H}(\mathcal{H}) \xrightarrow{\mathcal{H}} (\mathcal{H} - \mathcal{H}) \mathcal{H}(\mathcal{H}) / \mathcal{H}(\mathcal{H}) \right) \quad (5.3.2)$$

By Proposition 2.4.3, it is a direct summand of $\mathcal{H}(\mathcal{H})$ of rank one. It is stable under the $\mathcal{H}$ -action. Moreover, we have $(\mathcal{H} - \mathcal{H}) \mathcal{H}(\mathcal{H}) \subset \mathcal{H}(\mathcal{H})$ and $(\mathcal{H} + \mathcal{H}) \mathcal{H}(\mathcal{H}) = (\mathcal{H})$. Replacing $\mathcal{H}$ with $\mathcal{H}$, we can similarly construct a line bundle $\mathcal{H} \subset \mathcal{H}(\mathcal{H})$.

Definition 5.3.1. The line bundle of modular forms $\mathcal{H}$ on $\mathcal{H}$ is the invertible sheaf of $\mathcal{H}$ -modules with inverse

$$\mathcal{H}^{-1} = \mathcal{H}(\mathcal{H}(\mathcal{H}), \mathcal{H}).$$

Similarly, one can also define

$$\mathcal{H}^{-1} = \mathcal{H}(\mathcal{H}(\mathcal{H}), \mathcal{H}).$$

For any $\mathcal{H} \rightarrow \mathcal{H}$, we define the line bundle of modular forms $\mathcal{H}$ (resp, $\mathcal{H}$) over $\mathcal{H}$ by pull-back.

Remark 5.3.2. We define the line bundles of modular forms following the construction in [How19], which is different from that in [BHK⁺20], cf. [How19, Remark 3.5]. The terminology is motivated by their global counterpart on the unitary Shimura variety, see [BHK⁺20, Remark 2.4.1] for further explanation.

Remark 5.3.3. Let $\mathcal{H}$ be the rigid generic fiber of the RZ space. We have the Grothendieck–Messing (or Rapoport–Zink) crystalline morphism in [RZ96] (which extends to the relative in [FGL08]):

$$\mathcal{H} : \mathcal{H} \longrightarrow \mathcal{H} \quad (5.3.3)$$

where $\mathcal{H}$ is the rigid analytic flag variety parameterizing all codimension one subspaces of $(\mathcal{H} + \mathcal{H})$. The tautological line bundle on $\mathcal{H}$ then pull-back to a line bundle $\mathcal{H}$ over $\mathcal{H} = \mathcal{H}$. This is the rigid generic fiber of line bundle of modular forms:

$$\mathcal{H} = \mathcal{H} = \mathcal{H}.$$

It is the $\mathcal{H}$ -adic analog of the line bundle of modular forms over the complex hermitian symmetric domain of $\mathcal{H}(\mathcal{H} - \mathcal{H})$.

5.4. Special cycles on splitting RZ spaces. Given a framing object $(\text{🐞}, \text{🐜}, \text{🐜}, \text{🔔})$, we define the space of *special quasi-homomorphisms*

$$\text{🌸} = \text{🌸}(\text{🐞}) := \text{🔗}^\circ(\text{🌸}, \text{🐞}). \quad (5.4.1)$$

Then 🌸 is an 🐜 -dimensional 🌊 -vector space. It carries a natural nondegenerate $\text{🌊}/\text{🌊}$ -hermitian form 🐜 : for $\text{🌸}, \text{🍌} \in \text{🌸}$, the composite

$$\text{🌸} \xrightarrow{\text{🍌}} \text{🐞} \xrightarrow{\text{🔔}} \text{🐞} \xrightarrow{\text{🌸}} \text{🌸} \xrightarrow{\text{🔔}} \text{🌸} \xrightarrow{\text{🍌}} \text{🌸}$$

lies in $\text{🔗}^\circ(\text{🌸})$ and, hence, identifies with an element $\text{🐜}(\text{🌸}, \text{🍌}) \in \text{🌊}$ via the isomorphism

$$\text{🌊} : \text{🌊} \xrightarrow{\sim} \text{🔗}^\circ(\text{🌸}).$$

Definition 5.4.1. (i) For $\text{🌸} \in \text{🌸}$, define the (🌸 -Kudla–Rapoport) special cycle $\text{🌸}(\text{🌸})$ as the closed subfunctor of 🐞 , such that for any 🌊 -scheme 🌸 , $\text{🌸}(\text{🌸})(\text{🌸})$ is the set of isomorphism classes $(\text{🔗}, \text{🐞}, \text{🔔}, \text{🐜}, \text{🐜}, \text{🔔})$ such that the quasi-homomorphism

$$\text{🐜} \circ \text{🌸} \circ \text{🐜} : \text{🌸} \times \text{🌸} \rightarrow \text{🔗} \times \text{🌸}$$

extends to a homomorphism $\text{🌸} \times \text{🌸} \rightarrow \text{🔗}$. For any 🌊 -scheme 🌸 over 🐞 , we define the special cycle $\text{🌸}(\text{🌸})$ on 🌸 as the pull-back of $\text{🌸}(\text{🌸})$ on 🐞 .

(ii) For $\text{🌸} \in \text{🌸}$, define the (🌸 -Kudla–Rapoport) special cycle $\text{🌸}(\text{🌸})$ as the closed subfunctor of 🐞 , such that for any 🌊 -scheme 🌸 , $\text{🌸}(\text{🌸})(\text{🌸})$ is the set of isomorphism classes $(\text{🔗}, \text{🐞}, \text{🔔}, \text{🐜}, \text{🐜}, \text{🔔})$ such that the quasi-homomorphism

$$\text{🐜} \circ (\text{🔔} \times \text{🌸}) \circ \text{🌸} \circ \text{🐜} : \text{🌸} \times \text{🌸} \rightarrow \text{🔗} \times \text{🌸}$$

extends to a homomorphism $\text{🌸} \times \text{🌸} \rightarrow \text{🔗}$. For any 🌊 -scheme 🌸 over 🐞 , we define the special cycle $\text{🌸}(\text{🌸})$ on 🌸 as the pull-back of $\text{🌸}(\text{🌸})$ on 🐞 .

We generalize Howard’s work [How19] to our case. Let $\text{🌸} \subset \text{🌸}$ be any closed formal subscheme, and denote by $\text{🌸} \subset \text{🌸}$ its ideal sheaf. The square 🌸 is the ideal sheaf of a larger closed formal subscheme

$$\text{🌸} \subset \widetilde{\text{🌸}} \subset \text{🌸}$$

called the *first-order infinitesimal neighborhood* of 🌸 in 🌸 .

Let 🌸 be a 🌊 -scheme over 🐞 and let $\text{🌸} \in \text{🌸}$. We consider the first-order infinitesimal neighborhood

$$\begin{array}{ccccc} \text{🌸}(\text{🌸}) & \hookrightarrow & \widetilde{\text{🌸}(\text{🌸})} & \hookrightarrow & \text{🌸} \\ \downarrow & & \downarrow & & \downarrow \\ \text{🌸}(\text{🌸}) & \hookrightarrow & \widetilde{\text{🌸}(\text{🌸})} & \hookrightarrow & \text{🌸} \end{array}$$

of the special cycles and their pull-backs to 🌸 .

By the definition of $\text{🌸}(\text{🌸})$, the restriction of 🌸 and 🔗 over 🌸 to $\text{🌸}(\text{🌸})$ has a distinguished morphism of strict 🌊 -modules

$$\text{🌸}|_{\text{🌸}(\text{🌸})} \xrightarrow{\text{🌸}} \text{🔗}|_{\text{🌸}(\text{🌸})},$$

which further induces an 🌊 -linear morphism of vector bundles on $\text{🌸}(\text{🌸})$

$$\text{🐜}(\text{🌸})|_{\text{🌸}(\text{🌸})} \xrightarrow{\text{🌸}} \text{🐜}(\text{🔗})|_{\text{🌸}(\text{🌸})} \quad (5.4.2)$$

that preserves the Hodge filtrations. By (relative) Grothendieck–Messing theory, we have a canonical extension to $\widetilde{\text{🌸}(\text{🌸})}$

$$\text{🐜}(\text{🌸})|_{\widetilde{\text{🌸}(\text{🌸})}} \xrightarrow{\widetilde{\text{🌸}}} \text{🐜}(\text{🔗})|_{\widetilde{\text{🌸}(\text{🌸})}},$$

which does not preserve the Hodge filtrations anymore. In particular, along the commutative diagram it induces a nontrivial composite morphism

$$\widetilde{\text{🌸}} : \text{🌸}(\text{🌿})|_{\widetilde{\text{🌸}}(\text{🌸})} \longrightarrow \text{🌿}(\text{🌿})|_{\widetilde{\text{🌸}}(\text{🌸})} \xrightarrow{\text{🌿}} \text{🌿}(\text{🌿})|_{\widetilde{\text{🌸}}(\text{🌸})} \longrightarrow \text{🌿}(\text{🌿})|_{\widetilde{\text{🌸}}(\text{🌸})}, \quad (5.4.3)$$

where the first arrow and last arrow come from the Hodge exact sequences for 🌿 and 🌿 respectively.

Proposition 5.4.2. [How19, Proposition 4.1] *The morphism (5.4.3) factors through $\text{🌸}(\text{🌿})|_{\widetilde{\text{🌸}}(\text{🌸})}$. In particular, it can be viewed as a morphism of line bundles*

$$\text{🌿}(\text{🌿})|_{\widetilde{\text{🌸}}(\text{🌸})} \xrightarrow{\text{🌿}} \text{🌸}(\text{🌿})|_{\widetilde{\text{🌸}}(\text{🌸})}. \quad (5.4.4)$$

The Kudla–Rapoport divisor $\text{🌸}(\text{🌸})$ coincides with the zero locus of $\widetilde{\text{🌸}}$. $\square$

Definition 5.4.3. The section

$$\text{🌿}(\text{🌸}) \in \text{🌿}(\text{🌸})|_{\widetilde{\text{🌸}}(\text{🌸})}, \text{🌿}(\text{🌸})|_{\widetilde{\text{🌸}}(\text{🌸})}$$

determined by (5.4.4) is called the *obstruction to deforming* 🌸 . As we explained, $\text{🌸}(\text{🌸})$ is the closed subscheme of $\text{🌸}(\text{🌸})$ defined by $\text{🌿}(\text{🌸}) = \text{🌿}$.

Proposition 5.4.4. *For any nonzero $\text{🌸} \in \text{🌸}$, if the closed formal subscheme $\text{🌸}(\text{🌸}) \subset \text{🌸}(\text{🌸})$ (resp. $\text{🌸}(\text{🌸}) \subset \text{🌸}(\text{🌸})$) is nonempty, it is a Cartier divisor; that is to say, it is defined locally by a single nonzero equation.*

Proof. Let 🌸 be the local ring of $\text{🌸}(\text{🌸})$ at a point $\text{🌸} \in \text{🌸}(\text{🌸})$, and let $\text{🌸} \supset \text{🌸}$ be the ideals of 🌸 corresponding to $\text{🌸}(\text{🌸}) \subset \widetilde{\text{🌸}}(\text{🌸})$. The line bundle $\text{🌿}(\text{🌸})$ is free on $\text{🌸} = \text{🌿}(\text{🌸})$ and the obstruction to deforming 🌸 becomes an 🌸 -module generator

$$\text{🌿}(\text{🌸}) \in \text{🌸}/\text{🌸}.$$

It follows from Nakayama’s lemma that $\text{🌸} \subset \text{🌸}$ is a principal ideal, and it only remains to show that $\text{🌸} \neq \text{🌸}$. Recall from Remark 5.3.3 that we have the Grothendieck–Messing (or Rapoport–Zink) crystalline morphism

$$\text{🌿} : \text{🌸} \longrightarrow \text{🌸},$$

It follows from [RZ96, Proposition 5.17] that 🌿 is étale.

Suppose $\text{🌸} = \text{🌸}$, this implies that we may find an open subset $\text{🌸} \subseteq \text{🌸}$ such that $\text{🌸}(\text{🌸})|_{\text{🌸}} = \text{🌸}$, and $\text{🌸} \subseteq \text{🌸}$ determines an admissible open subset

$$\text{🌸} \subseteq \text{🌸}.$$

But the image $\text{🌿}(\text{🌸}) \subset \text{🌸}$ will factor through the rigid space associated to a closed subscheme of 🌸 , which is a contradiction. See [How19, Proposition 4.3] for the detailed argument (one can alternatively establish properness by using the 🌸 -adic uniformization of the integral model of the unitary Shimura variety).

The same argument works here for 🌸 -cycles by replacing 🌿 with 🌿 and 🌸 with 🌸 . In particular, we have that the closed formal subscheme $\text{🌸}(\text{🌸}) \subset \text{🌸}(\text{🌸})$ is a Cartier divisor when $\text{🌸} \neq \text{🌸}$. $\square$

Next, we conclude the linear invariance properties of special cycles on 🌸 .

Let 🌸 be a regular formal scheme. Recall from [Zha21a, Appendix B] that $\text{🌸}'(\text{🌸})$ is the free abelian group generated by symbols $[\text{🌸}]$, where 🌸 runs over all isomorphism classes of coherent 🌸 -modules, modulo the relations $[\text{🌸}] + [\text{🌸}] = [\text{🌸}]$ for every short exact sequence

$$\text{🌸} \longrightarrow \text{🌸} \longrightarrow \text{🌸} \longrightarrow \text{🌸} \longrightarrow \text{🌸}.$$

This group is endowed with a ring structure, where multiplication is defined via the derived tensor product.

Let $\mathbb{A}^1_{\mathbb{Q}} \rightarrow \mathbb{A}^1_{\mathbb{Q}} \times \mathbb{A}^1_{\mathbb{Q}}$ be a regular formal scheme over $\mathbb{A}^1_{\mathbb{Q}} \times \mathbb{A}^1_{\mathbb{Q}}$, for instance, we can take $\mathbb{A}^1_{\mathbb{Q}} = \mathbb{A}^1_{\mathbb{Q}} \times \mathbb{A}^1_{\mathbb{Q}} \hookrightarrow \mathbb{A}^1_{\mathbb{Q}} \times \mathbb{A}^1_{\mathbb{Q}}$ to be the splitting RZ space, this recovers the setting in Proposition 1.5.3.

When $\mathbb{A}^1_{\mathbb{Q}} \neq \mathbb{A}^1_{\mathbb{Q}}$, by Proposition 5.4.4, two special cycles $\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}})$ and $\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}})$ define classes $[\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}})]$ resp. $[\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}})]$ in the Grothendieck group $\mathcal{K}_0(\mathbb{A}^1_{\mathbb{Q}})$. We define $[\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}})]$, resp. $[\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}})]$, as the classes in $\mathcal{K}'_0(\mathbb{A}^1_{\mathbb{Q}})$ of the following complex of $\mathbb{A}^1_{\mathbb{Q}}$ -modules

$$(\cdots \rightarrow \mathbb{A}^1_{\mathbb{Q}} \rightarrow \mathbb{A}^1_{\mathbb{Q}} \rightarrow \mathbb{A}^1_{\mathbb{Q}} \rightarrow \mathbb{A}^1_{\mathbb{Q}} \rightarrow \mathbb{A}^1_{\mathbb{Q}}), \quad \text{resp.} \quad (\cdots \rightarrow \mathbb{A}^1_{\mathbb{Q}} \rightarrow \mathbb{A}^1_{\mathbb{Q}} \rightarrow \mathbb{A}^1_{\mathbb{Q}} \rightarrow \mathbb{A}^1_{\mathbb{Q}} \rightarrow \mathbb{A}^1_{\mathbb{Q}}),$$

cf. [How19].

Corollary 5.4.5. *Let $\mathbb{A}^1_{\mathbb{Q}}, \dots, \mathbb{A}^1_{\mathbb{Q}} \in \mathcal{K}_0$ and $\mathbb{A}^1_{\mathbb{Q}}, \dots, \mathbb{A}^1_{\mathbb{Q}} \in \mathcal{K}_0$ generate the same $\mathbb{A}^1_{\mathbb{Q}}$ -submodule. When $\mathbb{A}^1_{\mathbb{Q}}$ is regular,*

$$[\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}})] \cdot [\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}})] \cdots [\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}})] = [\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}})] \cdot [\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}})] \cdots [\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}})] \in \mathcal{K}'_0(\mathbb{A}^1_{\mathbb{Q}}).$$

Similarly, we have

$$[\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}})] \cdot [\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}})] \cdots [\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}})] = [\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}})] \cdot [\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}})] \cdots [\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}})] \in \mathcal{K}'_0(\mathbb{A}^1_{\mathbb{Q}}).$$

Proof. This is a purely homological algebra argument. By comparing with [How19, §5], we see that all we need to know are the following facts, which all hold in our situation:

- (1) $\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}}) \subset \mathbb{A}^1_{\mathbb{Q}}$ is a Cartier divisor when $\mathbb{A}^1_{\mathbb{Q}} \neq \mathbb{A}^1_{\mathbb{Q}}$. Hence locally $\mathbb{A}^1_{\mathbb{Q}}(\mathbb{A}^1_{\mathbb{Q}})$ can be cut by a single equation.
- (2) We can locally choose sections of a line bundle as obstructions to the first-order infinitesimal deformation, so that we can apply [How19, Lemma 5.2], which is a purely commutative algebra statement.
- (3) $\mathbb{A}^1_{\mathbb{Q}}$ is regular, this is used in [How19, (5.7)]. $\square$

Remark 5.4.6. The importance of including additional filtrations $\mathbb{A}^1_{\mathbb{Q}} \subset \mathbb{A}^1_{\mathbb{Q}}$ for both $\mathbb{A}^1_{\mathbb{Q}} = \mathbb{A}^1_{\mathbb{Q}}$ and $\mathbb{A}^1_{\mathbb{Q}} = \mathbb{A}^1_{\mathbb{Q}} - \mathbb{A}^1_{\mathbb{Q}}$ in the naive splitting model becomes evident now. This is because the proof of Proposition 5.4.4 relies on the line bundle of modular forms for both $\mathbb{A}^1_{\mathbb{Q}}$ -cycles and $\mathbb{A}^1_{\mathbb{Q}}$ -cycles. Namely, $\mathbb{A}^1_{\mathbb{Q}}$ is derived from the $\mathbb{A}^1_{\mathbb{Q}}$ -filtration and $\mathbb{A}^1_{\mathbb{Q}}$ is derived from the $\mathbb{A}^1_{\mathbb{Q}}$ -filtration, as outlined in Definition 5.3.1.

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